Physics-Informed AecroFormer for Stable Aerosol Microphysical Retrieval From Multiwavelength Lidar

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***Abstract*—Aerosol microphysical retrieval from multiwavelength lidar is an ill-posed inverse problem because limited $3\beta + 2\alpha$ optical observations must constrain multiple coupled aerosol properties through nonlinear Lorenz–Mie light-scattering physics. Small observational errors may therefore be amplified into unstable estimates of refractive index, particle size distribution, and concentration. Building on the AecroFormer backbone introduced in our previous work, this study develops a physics-informed and uncertainty-aware estimator framework for more stable aerosol microphysical retrieval under the established $3\beta + 2\alpha$ setting. The framework combines group-normalized multiwavelength optical inputs that retain Ångström-exponent-related spectral information with two explicit same-wavelength extinction-to-backscatter (lidar-ratio) features, observation-space physical consistency, dynamic non-gridded sampling, heteroscedastic uncertainty modeling, and physically guided data augmentation. The retrieval problem is formulated for fine-mode-dominated aerosols represented by a unimodal lognormal particle size distribution. Multi-seed synthetic experiments show that the proposed components play complementary and parameter-dependent roles: the reorganized input representation provides the most consistent improvement in point-estimate accuracy, while the full uncertainty-enabled configuration maintains competitive accuracy and provides parameter-dependent predictive intervals. Compared with a multivariate Full-permutAtion based Stochastic foresT (FAST)-based KNN lookup-table reference, the final configuration reduces the MAE of m_r from 0.0673 to 0.0494±0.0001, the MARE of r_{med} from 14.21% to 12.93%±0.05%, and the MARE of N from 86.65 to 51.58%±0.91% under the standard-noise setting. External validation using observations from the California deployment of NASA's Deriving Information on Surface Conditions from Column and Vertically Resolved Observations Relevant to Air Quality (DISCOVER-AQ) campaign, including 30 matched profiles from 8 observation days and 14 B-200 flight segments, yields an effective-radius MARE of 14.59%±1.67%, comparable to the LUT value of 14.36, while reducing the MAE of 532-nm single-scattering albedo from 0.0370 to 0.0247±0.0032. A full-day case on 1 February 2013 further demonstrates more continuous time–height and vertical-profile responses, together with explicit predictive intervals that identify altitude ranges of larger retrieval uncertainty. Mechanistic analyses based on noise degradation, empirical conditional distributions, and local perturbation-plane responseindicate that the performance gain of the proposed framework does not arise from additional observational information, but from reducing LUT-like candidate switching and producing more continuous and locally stable estimator responses. The advantage remains parameter dependent and bounded by observation quality, calibration accuracy, training-distribution coverage, and forward-model assumptions.**

***Index Terms*—Aerosol microphysical retrieval, Multiwavelength lidar, Ill-posed inverse problem, Physics-informed deep learning, Lookup table retrieval**

I. INTRODUCTION

Aerosols are an important component of the atmosphere. They are closely related not only to air quality and human health [1], but also to the Earth's atmospheric radiation balance through the scattering and absorption of solar radiation [2-5]. In addition, aerosols influence cloud formation and cloud microphysical processes through aerosol–cloud interactions [6-8] and can further modify precipitation and regional climate [9]. Therefore, accurate observation of aerosol optical properties and reliable retrieval of microphysical properties, including particle size distribution, complex refractive index, and particle concentration, are essential for atmospheric remote sensing and climate and environmental research [10]. Lidar provides vertically resolved aerosol observations with high spatial and temporal resolution. In particular, multiwavelength Raman lidar and high-spectral-resolution lidar (HSRL) provide aerosol backscatter and extinction profiles through different measurement principles [11-13], thereby supplying key optical inputs for subsequent

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aerosol microphysical inversion. HSRL exploits the different spectral broadening of molecular and particulate backscatter signals to separate their contributions, thereby enabling aerosol extinction and backscatter coefficients to be derived independently at the HSRL wavelengths without prescribing a lidar ratio (extinction-to-backscatter ratio at a given measurement wavelength). These optical products provide an important observational basis for subsequent aerosol microphysical inversion [14], [15]. Beyond aerosol applications, Zhang et al. [16] developed a dual-field-of-view HSRL algorithm for simultaneous profiling of water cloud optical and microphysical properties without relying on a subadiabatic assumption.

A relatively systematic theoretical and methodological framework has already been established for aerosol microphysical retrievals from multiwavelength lidar observations. Previous studies have shown that, under commonly used observational configurations, 3β+2α, i.e., three aerosol backscatter coefficients at 355, 532, and 1064 nm (β_{355}, β_{532}, and β_{1064}) and two aerosol extinction coefficients at 355 and 532 nm (α_{355} and α_{532}), can be regarded as the minimum information requirement for effective retrieval of aerosol microphysical properties for the case that the observed aerosol particles are spherical [17,18], while observational accuracy strongly affects retrieval stability and interpretability [19,20]. The NASA Langley HSRL-2 system is one instrumental implementation capable of providing this optical configuration, with aerosol backscatter coefficients at 355, 532, and 1064 nm and aerosol extinction coefficients at 355 and 532 nm. These HSRL-2-derived optical profiles have subsequently been used as inputs to aerosol microphysical inversion algorithms. For example, Sawamura et al. evaluated vertically resolved fine-mode number, surface-area and volume concentrations, and effective radius retrieved from HSRL-2 observations against 108 collocated airborne in situ profiles during the Deriving Information on Surface Conditions from Column and Vertically Resolved Observations Relevant to Air Quality (DISCOVER-AQ) campaign [21]. Such studies demonstrate the value of HSRL-derived 3β+2α optical products for constraining aerosol microphysical properties. However, the subsequent microphysical inversion remains ill posed because the five optical coefficients must still constrain multiple coupled aerosol parameters and therefore remains sensitive to observational uncertainty, forward-model assumptions, and the adopted inversion constraints [22]. On this basis, Müller and coworkers developed a series of classical approaches, including regularization-based inversion and lookup table methods [20,22-25], which laid the theoretical foundation for multiwavelength lidar microphysical retrieval and promoted its practical applications [26]. At the same time, these studies clearly showed that aerosol microphysical retrieval from multiwavelength lidar is essentially an ill-posed inverse problem under limited observational constraints [20]. In this problem, a finite-dimensional vector of lidar optical observations is used to infer multiple microphysical parameters whose contributions to the measured optical quantities are strongly coupled through the nonlinear light-scattering forward model [27]. Consequently, different combinations of refractive index, particle size, distribution width, and particle concentration may produce similar optical responses within observational uncertainty. Even small measurement errors can therefore be amplified in the retrieved parameter space, leading to unstable solutions, local non-uniqueness, and strong dependence on prior constraints.

Against this background, subsequent methodological approaches have largely focused on improving three aspects of aerosol microphysical retrieval under ill-posed conditions: retrieval stability, computational and engineering applicability, and uncertainty characterization for quality assurance. One representative approach is the development of fast retrieval frameworks based on lookup tables (LUT). Wang, Zhou, and others introduced strategies such as machine learning pruning and k-nearest-neighbor retrieval (KNN) into the traditional framework, substantially improving the computational efficiency of LUT-based methods while maintaining good applicability to real observations [28,29]. Among these methods, the multivariate Full-permutAtion based Stochastic foresT (FAST) method was selected as the LUT-based reference in this study because it has demonstrated competitive retrieval accuracy and computational efficiency for multiwavelength lidar aerosol microphysical inversion. Our implementation confirmed its competitive performance, providing a rigorous reference for evaluating the advantages and limitations of the proposed deep learning estimator. By transforming a complicated inversion into a candidate search and screening problem, such methods have clear practical advantages. However, the essence of LUT retrieval still lies in the matching between discrete parameter grids and observations [19], and its output remains constrained by the predefined sample library, parameter discretization scheme, and candidate partition strategy. Even when combined with local interpolation or nearest-neighbor search, the sensitivity of this class of methods to observational perturbations, grid structure, and local candidate switching is difficult to eliminate fundamentally [30,31]. Therefore, although LUT methods can greatly improve retrieval efficiency, they do not alter the intrinsic sensitivity of the problem itself to noise amplification and local instability under ill-posed conditions [13,20].

By contrast, deep learning (DL) offers a new possibility for this class of ill-posed retrieval problems [32,33]. By directly learning the nonlinear mapping from optical observations to microphysical parameters, DL has the potential to move beyond explicit discrete lookup tables and gain advantages in retrieval efficiency, fitting ability in continuous parameter space, and local smoothness of the response. In our previous work, AecroFormer, an abbreviation of AErosol miCROphysical transFORMER Framework, was introduced as a lightweight MLP–MHA-based model for end-to-end aerosol microphysical retrieval from multiwavelength Raman lidar observations [32]. That study established the AecroFormer backbone, evaluated the 3β+2α retrieval configuration, investigated noise-robust training through augmentation and perturbation tests, compared the learned estimator with LUT-based retrieval methods, and included a lidar–in situ validation case. Accordingly, the

present study does not claim novelty in the basic network architecture, the 3β+2α task definition, the general motivation of noise-robust training, the use of LUT comparison, or the inclusion of real-observation validation itself. However, most existing DL methods still rely on the supervised-learning paradigm, and their performance depends strongly on the training-sample distribution, input representation, definition of supervisory targets, and the quality and uncertainty characteristics of the optical observations [34,36]. For ill-posed aerosol retrieval, observational errors or input perturbations beyond the training distribution may lead to extrapolation bias, locally non-smooth responses, and unreliable estimates [17,34,36].

In parallel, previous studies have identified several relatively stable optical-microphysical relationships that can serve as important priors for constraining the solution space [27,37]. Robust DL retrieval therefore requires explicit physical consistency constraints, alongside parameter-space supervision, to suppress solutions incompatible with the Lorenz–Mie-based forward operator [27,37]. In other words, the key value of DL in this problem does not simply lie in replacing a traditional retrieval algorithm with a more complex network, but rather in whether it can absorb the physical constraints, statistical priors, and robust training mechanisms required by an ill-posed inverse problem while learning a continuous mapping. Deep-learning studies specifically targeting aerosol microphysical retrieval from multiwavelength lidar remain scarce. Our previous AecroFormer study established the feasibility of end-to-end retrieval and evaluated its accuracy, noise robustness, and performance relative to LUT-based methods [32], while related studies have examined learning-based inversion under limited information and observational uncertainty in broader atmospheric retrieval contexts [17], [35]. However, systematic analyses remain limited regarding why learned estimators behave differently under ill-posed conditions, for which parameters and perturbation regimes their advantages are most evident, and where these advantages begin to weaken.

The need to understand estimator behavior under limited observational constraints is not unique to aerosol microphysical retrieval from multiwavelength lidar. Similar non-uniqueness and error amplification occur in other atmospheric remote-sensing and Earth-system estimation problems, including joint cloud–aerosol retrieval [38], multichannel passive remote-sensing retrieval [39], parameter recovery under insufficient polarization or spectral constraints [40], and data-assimilation-based state estimation [41]. Although deep learning has shown substantial promise in atmospheric prediction [42,43], inverse retrieval poses distinct challenges. Limited observational information, nonlinear forward physics, and strong parameter coupling can amplify observational errors and introduce ambiguity into retrieved microphysical properties. These challenges motivate examining both the stability and the limitations of learned aerosol microphysical estimators. Aerosol microphysical retrieval from multiwavelength lidar therefore provides a representative case for examining how DL changes estimator behavior under ill-posed conditions, as well as where its benefits remain limited.

Against this background, this study extends the existing AecroFormer backbone into a physics-informed and uncertainty-aware estimator framework for more stable aerosol microphysical retrieval [32]. Rather than proposing a more complicated network architecture, the present work reconstructs the training framework around the principal constraints of the ill-posed inverse problem. Specifically, it combines a group-normalized multiwavelength optical representation that preserves spectral relationships across wavelengths, supplemented by two explicit same-wavelength lidar-ratio features, with an observation-space physical consistency constraint, dynamic non-gridded sampling, heteroscedastic uncertainty modeling, and physically guided data augmentation. Its applicability under selected real observational conditions is further assessed using collocated airborne in situ measurements. In addition, the differences between the proposed continuous estimator and FAST-based discrete candidate matching are examined from three complementary perspectives: statistical degradation, empirical conditional distributions, and local responses on observation perturbation planes.

Accordingly, this study is concerned not only with whether selected DL configurations can outperform FAST-based LUT retrieval, but also with three more fundamental questions. First, why can a learned continuous estimator exhibit greater stability in this ill-posed retrieval problem? Second, for which parameters and under which noise or perturbation conditions is this advantage most evident? Third, how do physical constraints and the associated training mechanisms affect point-estimate accuracy, observation-space consistency, and predictive credibility under synthetic and selected field conditions?

The remainder of this paper is organized as follows. Section II introduces the forward simulation, the retrieval framework, and the design of each training mechanism. Section III presents the simulation-based ablation experiments and the NASA external validation results which are based on data taken during the DISCOVER-AQ California field campaign. Section IV discusses the advantages, boundaries, and limitations of DL in ill-posed retrieval from the perspectives of perturbation analysis and noise analysis. Section V concludes the paper and provides an outlook.

II. Methods

This section introduces the retrieval framework adopted in this study and the associated training mechanisms. As shown in Fig. 1, AecroFormer is used as the backbone model, while its basic network architecture is kept unchanged. The training framework is systematically extended from three aspects: data processing, supervision form, and physical constraints. Specifically, on the data side, richer normalized input representations, physically guided data augmentation, and dynamic non-gridded sampling are introduced to improve generalization in continuous parameter space and alleviate the grid locking effect. On the model side, a heteroscedastic uncertainty output branch is added to characterize sample-

dependent predictive credibility. On the training side, parameter space supervision and observation space physical consistency constraints are jointly optimized. Overall, this study does not propose a new network backbone. Instead, it extends AecroFormer into a physics-informed learning framework that is better suited to ill-posed aerosol microphysical retrieval problems.

A. Forward simulation and retrieval task definition

This study focuses on aerosol microphysical retrieval for fine-mode-dominated conditions represented by a unimodal lognormal particle size distribution. This parameterization defines the scope of the retrieval problem considered here and should not be interpreted as a general description of ambient aerosol populations which usually are bimodal in the optically active range and the standard observation wavelengths of aerosol lidars

Although the model architecture and forward-model interface are structurally extensible to a bimodal state representation, a validated multimodal retrieval under the standard $3\beta + 2\alpha$ configuration involves substantially greater challenges in observability, forward-model fidelity, and independent validation. Within a controlled parameter space, the forward mapping from microphysical states to optical observables is constructed, and the corresponding inverse mapping is then learned. Compared with the bimodal case, the unimodal setting can greatly reduce the additional parameter coupling, compensation, and identifiability burden introduced by a bimodal parameterization. This choice is also motivated by the role of the mode width in the retrieval problem. Even in the unimodal case, the mode width $\boldsymbol{\sigma}$, mathematically expressed as the geometric standard deviation of the lognormal size distribution, controls the spread of the particle size distribution and is strongly coupled with the median radius in shaping the optical response. For bimodal particle size distributions, the complexity does not increase in a purely additive manner, because multiple mode widths, characteristic radii, and relative volume contributions can compensate each other in the optical parameter space.

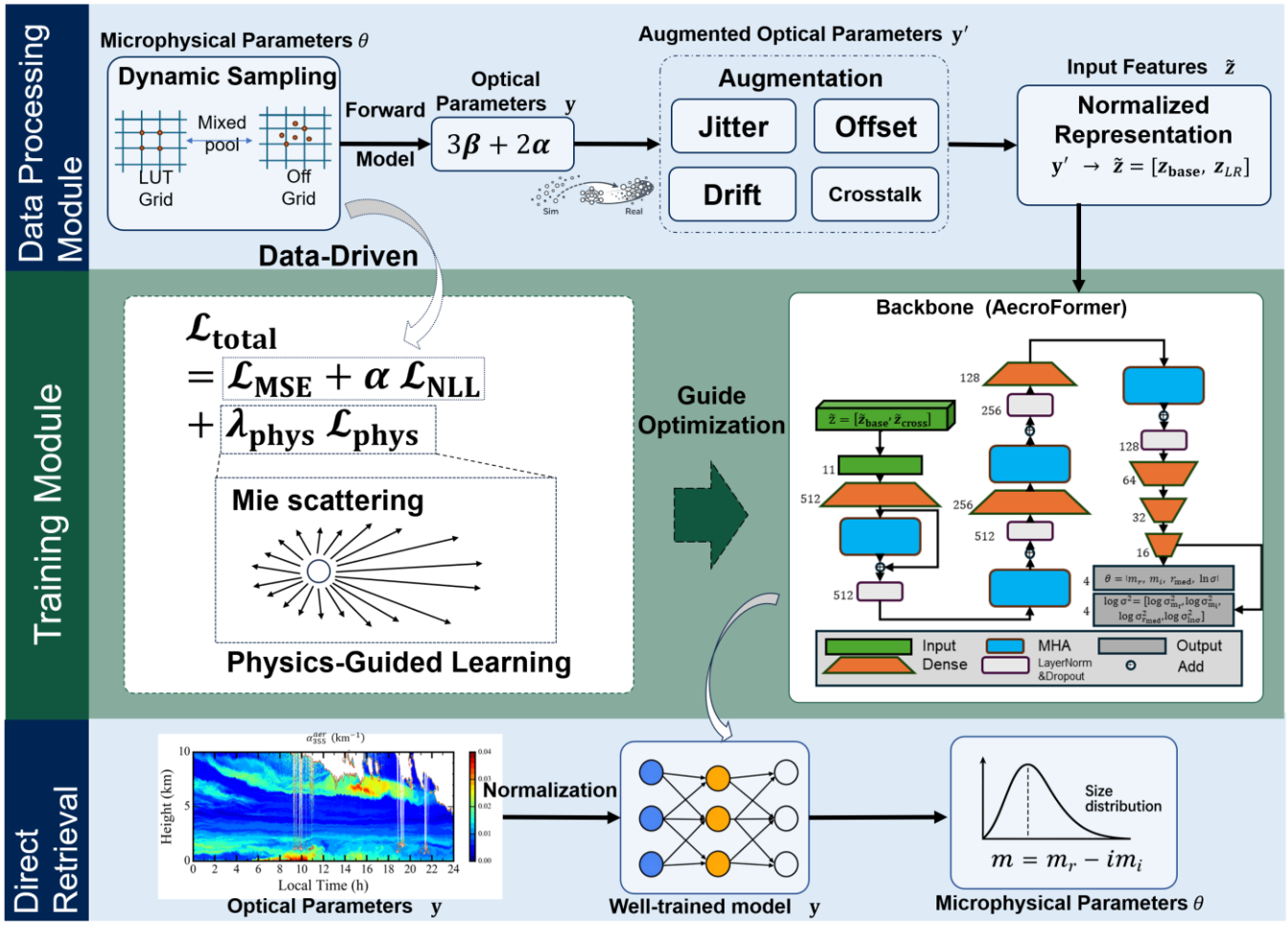


Fig. 1. Schematic diagram of the proposed physics-informed AecroFormer retrieval framework. The data-processing module combines dynamic sampling, forward simulation, optical augmentation, and input normalization. The training module jointly optimizes parameter-space supervision, heteroscedastic uncertainty modeling, and Lorenz–Mie-based physical consistency. The trained model then directly retrieves aerosol microphysical parameters from normalized lidar optical observations.

In this context, the ill-posedness of aerosol microphysical retrieval refers to the fact that the limited 3β+2α optical observations cannot uniquely determine the complete aerosol microphysical state. The forward mapping from microphysical parameters to lidar-measured optical quantities is nonlinear and highly coupled, so different combinations of complex refractive index, particle size (respectively particle size distribution), mode width, and particle number concentration may produce very similar optical responses within the observational uncertainty. Consequently, small perturbations in the optical observations can be amplified into large changes in the retrieved parameters, and the inverse solution may become locally non-unique or strongly dependent on the estimator. This property is the fundamental reason why retrieval stability, physical consistency, and estimator behavior under perturbations need to be examined in addition to conventional error metrics.

Throughout this paper, scalar quantities are denoted by italic symbols, vectors by bold lowercase symbols, and matrices by bold uppercase symbols. Sets are denoted using calligraphic or uppercase Greek symbols. Descriptive subscripts are written in upright type, whereas sample, target, and channel indices are italic. A hat denotes an estimated quantity, and $(\cdot)^{\top}$ denotes transpose.

Let the true microphysical parameter vector be denoted by $\boldsymbol{\theta}$, and the observation vector by $\mathbf{y}$. They satisfy the condition

$$\mathbf{y} = f(\boldsymbol{\theta}), \tag{1}$$

where $f(\cdot)$ denotes the forward operator based on the physical principles of light scattering. In this study, the microphysical parameter vector to be retrieved is defined as

$$\boldsymbol{\theta} = [m_r,\, m_i,\, r_{med},\, \ln\, \sigma]^{\top}, \tag{2}$$

where m_r and m_i denote the real and imaginary parts of the complex refractive index, respectively, r_{med} denotes the volume median radius of the investigated particle size distribution, and σ denotes the mode width of the lognormal size distribution, commonly expressed as the geometric standard deviation. In view of the limited lidar optical constraints, the inverse problem is highly ill-posed, and the estimation of the particle number concentration N is particularly sensitive to retrieval instability and scaling ambiguity. This is partly because the lidar wavelengths used here do not constrain the complete particle size distribution uniformly, but are most sensitive to a finite, wavelength- and refractive-index-dependent optically active size range. To reduce this effect and focus the comparison on size-distribution-related microphysical parameters, the total volume concentration is fixed at $V_t = 1.0\ \mu\mathrm{m}^3\ \mathrm{cm}^{-3}$ in the forward simulation, and N is not treated as an independent retrieval variable during training.

This treatment follows an established dimensionality-reduction strategy used in LUT-based and learning-based aerosol microphysical retrievals. By separating the size-distribution shape and refractive-index parameters from the absolute aerosol-loading scale, it reduces the number of simultaneously retrieved variables and partly alleviates the ill-posedness of the inverse problem. It does not imply that the aerosol volume concentration is fixed in real observations. After the size-distribution and refractive-index parameters have been retrieved, the concentration scale can be recovered from the absolute magnitude of the measured optical quantities.

For this parameterization, the particle size distribution is assumed to follow a unimodal lognormal form, jointly determined by r_{med} and σ. Given the fixed total volume concentration V_t, the corresponding particle number concentration N can be derived, and the complete unimodal particle size distribution $n(r)$ is then determined. Based on this forward formulation, additional microphysical quantities can be derived. In particular, the effective radius is defined as

$$r_e = \frac{\int r^3 n(r)\, dr}{\int r^2 n(r)\, dr}, \tag{3}$$

which will be used later on as an important quantity in the external validation of the retrieval results. The integration is performed over particle radii of 0.0005–15 μm using 1000 logarithmically spaced grid points.

Within this forward framework, the corresponding optical observables are calculated using Lorenz-Mie scattering theory. Specifically, for a given θ, Lorenz–Mie theory is used to calculate the single-particle extinction cross section and the 180° backscatter cross section at each lidar measurement wavelength as functions of particle radius and complex refractive index. These radius-dependent quantities are then weighted by the unimodal lognormal particle size distribution and integrated over radius using a precomputed high-resolution Mie lookup table:

$$\mathbf{y} = [\alpha_{355}, \beta_{355}, \alpha_{532}, \beta_{532}, \beta_{1064}]^{\top} \tag{4}$$

Here, α denotes the extinction coefficient and β denotes the backscatter coefficient, i.e., exactly 180° in the case of lidar observations. The backscatter quantities include the 4π steradian normalization and are uniformly converted to the km^{-1} scale which is typically used as the height scale in lidar observations.

According to the practical distribution of aerosol microphysical properties [32], the parameter space used in the main experiments is set as follows: $m_r \in [1.30, 1.70]$ with a step of 0.02; $m_i \in 0 \cup [10^{-6}, 0.5]$, where the nonzero part is sampled in logarithmic space with a total of 501 values; $ln\sigma \in [0.38, 0.50]$ with a step of 0.01; and $r_{med} \in [50, 500]$ nm with a step of 10 nm. Based on this discrete parameter space, a forward lookup table is precomputed, resulting in 6,291,558 parameter observation samples, which are used as the static lookup table training baseline. Furthermore, continuous random sampling is introduced within the same parameter ranges to generate dynamic training samples in real time, so as to support comparisons among different training data construction strategies at a later stage.

It should be emphasized that both the static LUT samples and the dynamic samples are generated using the same forward computation process. This forward process is also used consistently for training data generation, synthetic validation

set construction, and subsequent consistency-based evaluation, thereby ensuring physical consistency and comparability across all experimental stages.

These assumptions define the intended operating domain of the present framework. It is primarily applicable to fine-mode-dominated, approximately spherical aerosols that can be represented by a unimodal lognormal particle size distribution. Strongly nonspherical particles, such as mineral dust, may introduce systematic errors because of the mismatch with the spherical Lorenz–Mie kernels. Coarse-mode-dominated populations may be represented incorrectly as an enlarged or broadened unimodal distribution. Complex internally or externally mixed particles and strongly absorbing aerosols may not be adequately represented by a single homogeneous, wavelength-independent complex refractive index. Moreover, bimodal or multimodal populations introduce additional mode radii, widths, and relative contributions that cannot generally be constrained reliably by standard $3\beta + 2\alpha$observations. Under these conditions, the retrieved result should be interpreted as an effective solution within the assumed model class rather than as a unique physical decomposition.

B. Retrieval framework

AecroFormer is a deep learning model developed for aerosol microphysical retrieval from multiwavelength lidar observations. Its basic idea is to incorporate a multi-head attention mechanism into a lightweight perceptron backbone, so as to learn the nonlinear mapping from multichannel optical observations to microphysical parameters. The original AecroFormer is trained in a supervised manner using sample pairs generated by forward simulation and performs end-to-end regression from optical observations to microphysical state variables under the typical 3β+2α configuration [32]. Its overall network architecture is illustrated in Fig. 1.

For input representation, the original AecroFormer uses a small set of ratio-based optical features to characterize the spectral relationships among the lidar channels. In the present study, the AecroFormer backbone is retained, while the optical input is reorganized to separate the spectral structure within the backscatter and extinction groups. Specifically, the five-dimensional optical observation vector $\mathbf{y}$ defined in Eq. (4) is used as the starting point. The three backscatter channels and the two extinction channels are normalized separately using their respective L_2 norms:

$$b_{\mathrm{norm}} = \sqrt{\beta_{355}^2 + \beta_{532}^2 + \beta_{1064}^2}, \qquad a_{\mathrm{norm}} = \sqrt{\alpha_{355}^2 + \alpha_{532}^2}. \tag{5}$$

Accordingly, the normalized basic optical feature vector is written as

$$\mathbf{z}_{\mathrm{base}} = \left[\frac{\beta_{355}}{b_{\mathrm{norm}}}, \frac{\beta_{532}}{b_{\mathrm{norm}}}, \frac{\beta_{1064}}{b_{\mathrm{norm}}}, \frac{\alpha_{355}}{a_{\mathrm{norm}}}, \frac{\alpha_{532}}{a_{\mathrm{norm}}}\right]^{\mathsf{T}}. \tag{6}$$

This group-wise normalization preserves the wavelength-dependent structure within the backscatter and extinction groups while reducing the influence of the absolute aerosol-loading scale. However, it removes the relative amplitude relationship between extinction and backscatter. To restore this physically important coupling, two same-wavelength lidar-ratio features are introduced:

$$\mathbf{z}_{\mathrm{LR}} = \left[\frac{\alpha_{355}}{\beta_{355}}, \frac{\alpha_{532}}{\beta_{532}}\right]^{\mathsf{T}}. \tag{7}$$

The final input feature vector is defined as

$$\mathbf{z} = [\mathbf{z}_{\mathrm{base}}^{\mathsf{T}}, \mathbf{z}_{\mathrm{LR}}^{\mathsf{T}}]^{\mathsf{T}}. \tag{8}$$

Six extinction–backscatter cross-ratio features have been widely adopted in FAST and related LUT-based retrieval developments to characterize the coupling between the extinction and backscatter channels [28,29]. They therefore constitute a physically meaningful and well-established candidate representation. However, their necessity as explicit neural-network inputs was further examined here because the group-normalized components in $\mathbf{z}_{\mathrm{base}}$ already retain most of the within-group spectral relationships. The input-ablation experiment presented in Supplementary Section E compares the normalized base representation, the addition of the two same-wavelength lidar ratios, and the addition of all six extinction–backscatter cross ratios. The results show that the two same-wavelength lidar ratios recover the principal benefit of the extinction-backscatter coupling representation, whereas the full six-ratio set provides no consistent additional improvement. The final input therefore retains only α_{355}/β_{355} and α_{532}/β_{532}, which have direct lidar-ratio interpretations while reducing feature redundancy and the propagation of uncertainty from cross-wavelength combinations, particularly those involving the relatively weak β_{1064} channel.

At the output side, the retrieved state is uniformly defined as the four-dimensional microphysical parameter vector θ, corresponding to the real part of the complex refractive index, the imaginary part of the complex refractive index, the volume median radius, and the logarithm of the mode width, $\ln\sigma$. This definition is consistent with the parameterization used in the forward simulation, so that the network output can be directly fed into the Lorenz–Mie-based forward operator $f(\theta)$ to reconstruct the corresponding optical observations and to support the subsequent physical consistency constraint. The retrieval model can thus be written as

$$\hat{\boldsymbol{\theta}} = g(\mathbf{z}), \tag{9}$$

where $\hat{\theta}$ denotes the retrieved estimate of the true microphysical parameter vector θ and $g(\cdot)$ denotes the nonlinear mapping implemented by AecroFormer or its extended configurations. For the uncertainty-enabled configurations, the model outputs both the predicted parameter values and their corresponding log variances, to support Gaussian negative log likelihood-based supervised training and predictive interval estimation. The details are given in Section II.E.

C. Physical consistency loss

Based on the retrieval framework explained above, a physical consistency loss is further introduced by explicitly

incorporating the Lorenz–Mie-based forward operator that maps microphysical parameters to lidar optical observables [34,44]. When the inverse mapping is approximated solely through supervised learning, the optimization target is usually to minimize the discrepancy between the predicted microphysical parameters and the ground-truth microphysical labels. In a physical inversion, however, the retrieved state is ultimately expected to reproduce the observations, which corresponds to minimizing the discrepancy between the observed and forward-calculated optical quantities. For an ill-posed inverse problem such as aerosol microphysical retrieval from multiwavelength lidar, a small local error in parameter space does not necessarily imply a small discrepancy in observation space. In particular, when the optical input is perturbed and the coupling among parameters is strong, the model may produce a solution that is numerically close to the label (microphysical parameters) but cannot remain consistent with the input optical observations after forward calculation, thereby weakening the consistency between the retrieved state and the observed optical response. For this reason, in addition to parameter-space supervision, a physical consistency loss is introduced so that the network output is constrained jointly in both parameter space and observation space.

The total loss function is written as

$$L = L_{\mathrm{sup}} + \lambda_{\mathrm{phys}} L_{\mathrm{phys}}, \tag{10}$$

where L_{sup} is the supervised loss, L_{phys} is the physical consistency loss, and λ_{phys} is the weight of the physical term. A sensitivity comparison was conducted using $\lambda_{\mathrm{phys}} = 0.3$, 1.0, and 3.0. Based on the overall balance between parameter-space accuracy and observation-space consistency, $\lambda_{\mathrm{phys}} = 1.0$ was adopted as the empirical setting in this study, rather than being interpreted as a universally optimal value. The physical constraint is introduced in a joint-training manner, i.e., as an additional observation-space consistency term on top of the parameter-space supervision loss.

The supervision term is defined in parameter space. For configurations without heteroscedastic uncertainty modeling, the network directly outputs the predicted microphysical parameter values, and the supervised loss is calculated using the mean squared error. For the uncertainty-enabled configurations, the supervision term is extended to a Gaussian negative log-likelihood loss, in which the network predicts both the mean value and the log variance of each microphysical parameter. This uncertainty-aware formulation is intended to represent the predictive uncertainty of the retrieved parameters, which may arise from measurement perturbations in the input lidar optical quantities and from the ill-posed nature of the inverse mapping. Regardless of the specific supervision form, L_{sup} provides the basic parameter-space constraint for learning the mapping from the five lidar optical observables $(\alpha_{355}, \beta_{355}, \alpha_{532}, \beta_{532}, \beta_{1064})$ to the target microphysical parameters.

The physical term L_{phys} is defined in the observation space. Specifically, let the network prediction be $\widehat{\boldsymbol{\theta}}$. The predicted parameter vector is then fed into the Lorenz–Mie-based forward operator $f(\cdot)$ described in Section II.A to reconstruct the corresponding optical observation vector. For fixed total volume concentration $V_t = 1.0\ \mu\mathrm{m}^3\mathrm{cm}^{-3}$, the corresponding optical observation vector is reconstructed as

$$\hat{\mathbf{y}} = f(\widehat{\boldsymbol{\theta}}). \tag{11}$$

The reconstructed optical vector $\hat{\mathbf{y}}$ is then directly compared with the original optical observation vector $\boldsymbol{y}$ of the considered measurement sample, thereby imposing a closed-loop constraint on observation-space physical consistency.

The differentiable forward operator does not directly differentiate through the Lorenz–Mie series calculation. Instead, high-resolution single-particle extinction and backscatter kernels are precomputed on a fixed grid of particle radius, m_r, and m_i, and stored as constant tensors during training. For each predicted $(\hat{m}_r, \hat{m}_i)$, the corresponding continuous optical kernels are obtained through multilinear interpolation implemented using torch.nn.functional.grid_sample. The predicted $\hat{r}_{\mathrm{med}}$ and $\hat{\sigma}$ are then used to construct the lognormal particle size distribution on the fixed radius grid, and the extinction and backscatter coefficients are reconstructed through trapezoidal integration.

Because the interpolation, particle-size-distribution calculation, and numerical integration are implemented entirely as differentiable PyTorch tensor operations, gradients of L_{phys} can propagate through $\hat{\mathbf{y}} = f(\widehat{\boldsymbol{\theta}})$ to the predicted microphysical parameters and network weights. Over the parameter domain considered in this study, the maximum relative difference between the interpolated operator and direct Lorenz–Mie calculations is below 0.1%, and the automatically differentiated gradients agree with finite-difference estimates in both direction and magnitude.

In its specific form, a hybrid optical consistency loss is adopted. Let $\hat{y}_c$ and y_c denote the predicted and observed values of the c_{th} measurement channel, respectively. A relative error term is first defined as

$$L_{\mathrm{rel}} = \frac{1}{C} \sum_{c=1}^{C} \frac{|\, \hat{y}_c - y_c \,|}{\max\left(|\, y_c \,|, \epsilon_{\mathrm{rel}}\right)}, \tag{12}$$

where C is the number of observation channels, and ϵ_{rel} is a fixed numerical floor which is used to avoid unstable amplification of the relative error when the optical value of a channel approaches zero. In this study, ϵ_{rel} is set to 10^{-12} for all channels. This value is not a physical parameter or a theoretically optimized regularization coefficient, but a numerical stabilizer introduced to prevent division by zero or near-zero denominators. Since the optical quantities used in the loss are several orders of magnitude larger than 10^{-12} in physically meaningful cases, this term has negligible influence except for numerically degenerate near-zero values. To further reduce the imbalance caused by the high dynamic range of the optical quantities, the corresponding logarithmic error term is defined as

$$L_{\log} = \frac{1}{C} \sum_{c=1}^{C} [\log(1 + \hat{y}_c) - \log(1 + y_c)]^2. \tag{13}$$

The final physical loss is written as

$$L_{\mathrm{phys}} = \eta L_{\mathrm{rel}} + (1 - \eta) L_{\log}, \tag{14}$$

where $\eta \in [0,1]$ is the mixing weight that controls the relative contribution of the relative-error and log-domain consistency terms. In this study, $\eta = 0.5$ was adopted as the final empirical setting after preliminary sensitivity experiments, giving equal weight to the two consistency terms. This value should be regarded as a practical loss-balancing choice rather than a theoretically prescribed constant. In the current implementation, this loss also supports smooth reweighting when the relative error exceeds a prescribed threshold, so that the training process pays more attention to samples with stronger observation-space inconsistency.

The purpose of L_{phys} is to provide an additional observation-space regularization by penalizing disagreement between the optical response reconstructed from the predicted state and the input observations. It is not intended to uniformly reduce the error of every retrieved microphysical parameter. Because the inverse mapping is intrinsically ill-posed, improved observation-space closure may coexist with unchanged or slightly increased errors for individual parameters. Its contribution is therefore evaluated separately in terms of parameter-space errors, selected bias structures, and forward-reconstruction residuals. A detailed field-based evaluation of the observation-space effect is provided in Supplementary Section D.

D. Dynamic non-gridded sampling

Traditional LUT methods and the supervised training schemes derived from them are usually built on discrete parameter grids. Even if an explicit lookup procedure is replaced by a neural network at the inversion stage, as long as the training samples are entirely drawn from a fixed sample set generated once in advance, the model still mainly learns an approximate mapping defined on a discretely sampled distribution rather than a stable inverse mapping in continuous parameter space [45,46]. For the ill-posed microphysical retrieval problem considered here, such grid-based training can easily make the model implicitly adapt to the predefined sampling structure, thereby weakening its generalization ability on off-grid samples or under continuous parameter perturbations.

To reduce this effect, a dynamic non-gridded sampling mechanism is introduced with reference to the parameterization used in FAST [28]. Specifically, the training samples are no longer composed solely of precomputed lookup table samples. Instead, they are constructed by combining static LUT sampling with continuous random sampling, so that the stability of the training dataset is retained while non-gridded samples are continuously introduced.

Let the total number of samples used in each training round be N, and let the proportion of dynamically generated random samples be r. In this study, r=0.5 is used as a fixed empirical setting throughout training, so that each training round contains an equal mixture of static lookup-table samples and dynamically sampled non-gridded samples. The value of r is not adaptively updated or learned during training; instead, new dynamic samples are generated in each round under the same sampling proportion. In this way, the training process repeatedly samples from the same bounded continuous parameter domain, providing progressively richer coverage of off-grid regions without attempting to construct an exhaustive continuous lookup table. This setting was adopted to balance the coverage and reproducibility of the static LUT samples with the increased parameter-space continuity provided by dynamic sampling. Very small values of r would make the training distribution close to the original gridded LUT and reduce the benefit of non-gridded sampling, whereas very large values of r would weaken the anchoring effect of the precomputed LUT samples and may increase training variability. Therefore, $r = 0.5$ was used as a balanced empirical choice rather than as a theoretically optimized constant

$$B_{\mathrm{rand}} = rB, B_{\mathrm{lut}} = (1-r)B \tag{15}$$

where B_{lut} is obtained by uniform sampling with replacement from the precomputed LUT, and B_{rand} is generated by resampling in continuous parameter space.

The dynamic samples are generated within the same parameter ranges defined in Section II.A for m_r, m_i, r_{med}, and $\ln\sigma$. Specifically, m_r, r_{med}, and $\ln\sigma$ are uniformly sampled within their respective domains, whereas m_i is sampled using a mixed strategy consisting of a point mass at zero and a logarithmically uniform distribution for the nonzero part. This strategy preserves both non-light-absorbing cases and absorbing cases spanning several orders of magnitude. Although the effective radius r_e is not independently sampled, it is determined by the sampled r_{med} and σ and is later used as a derived quantity for external validation.

For each randomly sampled parameter vector $\boldsymbol{\theta}$, the corresponding five-dimensional optical observation vector $\mathbf{y}$ is calculated using the same Lorenz–Mie-based forward operator $f(\boldsymbol{\theta})$, and the feature vector $\mathbf{z}$ is then constructed according to Section II.B. Thus, the dynamic samples are not restricted to the discrete support of the LUT but cover intermediate regions in continuous parameter space. Since new samples are independently generated in each training round from predefined distributions, the method avoids a fixed clustering pattern in a specific region of the parameter space, although small random fluctuations may occur within individual batches.

During training, the training set is therefore not treated as a fixed sample collection, but rather as a mixed distribution that is continuously updated throughout the training process. The key purpose is to avoid over adaptation to a limited discrete grid and to encourage the network to learn a more continuous and smoother inverse mapping in parameter space. For the ill-posed microphysical retrieval problem considered here, this dynamic data mechanism can continuously introduce new off-grid samples while keeping the physical parameter boundaries consistent, thereby alleviating the distribution locking effect caused by fixed lookup table sampling and improving model generalization to continuous perturbations, out-of-sample shifts, and locally unstable regions.

E. Heteroscedastic uncertainty modeling

Aerosol microphysical retrieval from multiwavelength lidar exhibits clear sample heterogeneity. The forward sensitivity of the optical observables to different microphysical parameters, the coupling strength among extinction and backscatter channels at different wavelengths, and the local degree of ill-posedness of the inverse problem all vary across different regions of the parameter space. As a result, different samples can differ substantially in retrieval difficulty and effective observability. Traditional point estimate regression assumes that all samples share the same noise structure and the same error scale, which is difficult to reconcile with the sample-dependent uncertainty encountered in the actual problem. For this reason, heteroscedastic uncertainty modeling is introduced in some model configurations [47], allowing the model to output both the predictive mean and the predictive variance for each sample and each target variable.

For the ith sample, let $\theta_i^{std} \in \mathbb{R}^d$ denote the standardized target vector, where $d = 4$ and the four components correspond to m_{r}, m_{i}, r_{med}, and $\ln \sigma_{\mathrm{g}}$, respectively. The uncertainty-enabled output head produces a predictive-mean vector $\boldsymbol{\mu}_i$and a log-variance vector $\mathbf{s}_i$:

$$g_{\mathrm{UNC}}(\mathbf{z}_i) = (\boldsymbol{\mu}_i, \boldsymbol{s}_i), \boldsymbol{\mu}_i, \mathbf{s}_i \in \mathbb{R}^d. \tag{16}$$

Here, μ_{ij} and s_{ij} denote the predicted mean and log variance, respectively, for the jth target variable of the ith sample. The corresponding predictive variance v_{ij} and standard deviation u_{ij} are

$$v_{ij} = \exp(s_{ij}), u_{ij} = \exp\left(\frac{s_{ij}}{2}\right) \tag{17}$$

During supervised learning, a hybrid loss-combining mean squared error and Gaussian negative log likelihood are adopted:

$$L_{\mathrm{sup}} = L_{\mathrm{MSE}} + \lambda_{\mathrm{NLL}} L_{\mathrm{NLL}}, \tag{18}$$

where λ_{NLL}controls the contribution of the heteroscedastic Gaussian NLL term. Values of 0.05, 0.1, 0.2, and 0.5 were examined, and the retrieval results showed limited sensitivity within this range. A very small value provides insufficient supervision for the variance branch, whereas an excessively large value may overemphasize uncertainty fitting at the expense of the predictive mean. Therefore, $\lambda_{\mathrm{NLL}} = 0.1$ was adopted as a balanced empirical setting rather than a universally optimal value. The MSE term in Eq. (18) is defined as

$$L_{\mathrm{MSE}} = \frac{1}{Bd} \sum_{i=1}^{B} \sum_{j=1}^{d} \left(\mu_{ij} - \theta_i^{std}\right)^2, \tag{19}$$

where N is the number of samples and d is the target dimension. This term constrains the basic fitting accuracy of the predictive mean to the target parameters. Under the assumption of independent Gaussian outputs, the Gaussian negative log-likelihood (NLL) term is written as

$$L_{\mathrm{NLL}} = \frac{1}{Bd} \sum_{i=1}^{B} \sum_{j=1}^{d} \left[\frac{(\mu_{ij} - \theta_i^{std})^2}{2 s_{ij}^2} + \frac{1}{2} \log s_{ij}^2\right] \tag{20}$$

In practical implementation, a weak regularization term may also be added to suppress unconstrained divergence of the variance branch. During training, the NLL term is computed for each sample and each target variable in the standardized parameter space, then averaged over samples and variables, and finally combined with L_{MSE} using the weight α to form the supervised loss. Because all target variables are standardized before loss calculation, the different numerical ranges of m_r, m_i, r_{med}, and $\ln \sigma$ do not directly dominate the relative contributions of the variable-wise NLL terms.

It should be noted that all supervision terms are computed in the standardized parameter space. In other words, both the network outputs μ and $log\sigma^2$ correspond to standardized target variables. Therefore, L_{MSE} and L_{NLL} are both defined on a consistent standardized scale during training.

With this design, the learning objective is no longer limited to a single-point estimate but includes both the predicted parameter value and its associated uncertainty level. Consequently, the network can adaptively represent sample- and variable-dependent retrieval difficulty, that is, the difficulty of fitting different microphysical parameters under different local optical constraints. This reduces the excessive influence of highly ill-posed samples on the overall optimization and provides a more natural statistical basis for subsequent joint training with physical constraints.

F. Physically guided data augmentation

The results of the original AecroFormer showed that appropriate noise augmentation can improve model robustness to noise. However, if the augmentation strategy consists only of independent Gaussian perturbations with uniform amplitude, the resulting error structure remains close to idealized white noise and is insufficient to characterize the more complex error sources encountered in real multiwavelength lidar observations. In practice, observational errors are often not simply independent. They may simultaneously include overall gain variations, channel-correlated perturbations, heteroscedastic noise, and low-probability anomalies. Moreover, different lidar channels may have different uncertainty characteristics. For instance, backscatter coefficients at stronger signal channels, such as 532 nm, are often better constrained than those at weaker long-wavelength channels such as 1064 nm, while extinction coefficients can be more sensitive to noise propagation and smoothing choices than backscatter coefficients. This issue becomes particularly relevant when a weak or more uncertain channel contributes to normalized features or ratio-based representations, because its uncertainty may propagate to several derived input quantities. These differences motivate the use of channel-dependent and physically guided perturbations rather than uniform independent Gaussian noise. Therefore, for multiwavelength lidar retrieval, it is necessary to model error sources in a more structured random manner according to channel characteristics and error propagation behavior to reduce the gap between simulated and real observations [45,46,48]. Based on this consideration, a physically guided data augmentation strategy

is introduced during training to impose perturbations on the input optical quantities that better reflect the characteristics of real observations.

Specifically, data augmentation is not treated here as simple noise injection, but as an approximate representation of realistic lidar observational error structures. Compared with a single independent Gaussian perturbation, this strategy incorporates overall gain changes, wavelength-group correlated errors, channel crosstalk, and low-probability anomalies [48], thereby constructing compound error patterns that are closer to real lidar observations during training. The purpose is to encourage the model to learn a more stable inverse mapping under multi-source perturbations, rather than being robust only to idealized white noise. For the ill-posed retrieval problem studied here, this physically guided augmentation does not eliminate the intrinsic non-uniqueness caused by limited optical constraints. Instead, it helps the network learn a more robust estimator within the range of observation uncertainties represented during training, thereby reducing the discrepancy between simulated and real observations.

In summary, this study does not alter the basic AecroFormer retrieval backbone. Instead, it systematically extends the training mechanism around generalization, stability, and physical consistency under ill-posed inverse conditions, thereby forming a physics-informed learning framework that is more suitable for aerosol microphysical retrievals from multiwavelength lidar. In this framework, dynamic non-gridded sampling alleviates the structural bias caused by discrete grid training at the sample distribution level. Heteroscedastic uncertainty modeling characterizes differences in retrieval difficulty across samples at the level of statistical supervision. The physical consistency loss explicitly introduces the Lorenz–Mie-based forward operator into the optimization objective and thereby suppresses infeasible solutions which are inconsistent with the observations. Finally, physically guided data augmentation improves the stability of the learned estimator by introducing representative observational uncertainty structures during training. The relevant training hyperparameters and augmentation settings are listed in Appendix A, and the effectiveness of these modules and strategies will be further examined in the validation and ablation experiments in Section III .

III. Ablation Experiments and Algorithm Validation

A. Ablation experiments on synthetic data

1) Experimental design and compared configurations

To evaluate the effects of the proposed strategies under controlled conditions, a set of ablation experiments was conducted for the same unimodal $3\beta + 2\alpha$ retrieval task. The Lorenz–Mie forward parameterization, microphysical parameter domain, AecroFormer backbone, training and test protocols, and evaluation metrics were kept unchanged across the deep-learning configurations. Only the optical input representation and the associated training components were varied, including the physical consistency constraint, dynamic non-gridded sampling, heteroscedastic uncertainty modeling, and data augmentation. This design enables the individual and combined effects of these components to be distinguished without introducing differences in network capacity or retrieval-task definition.

As summarized in Table I, nine configurations were considered. ORI represents the published AecroFormer setting and retains its original input representation and conventional augmentation strategy. ORI-P adds the physical consistency constraint while preserving the original input setting, allowing the effect of the physical term to be evaluated independently of the reorganized input representation. BAS denotes the unified baseline developed in this study, while BAS-P and BAS-D independently introduce the physical consistency constraint and dynamic non-gridded sampling, respectively. DYN combines these two components, UNC further incorporates heteroscedastic uncertainty modeling, and NOA removes data augmentation from UNC to evaluate its contribution under otherwise unchanged conditions. These configurations form a set of matched comparisons rather than a strictly progressive sequence in which every additional module is expected to improve every metric.

LUT denotes the FAST-based KNN lookup-table retrieval [28, 29] and serves as the traditional retrieval reference. It uses the reorganized optical representation adopted in the controlled comparison and retrieves microphysical states by matching the observations against a precomputed discrete physical state space. FAST was selected because previous studies have demonstrated its competitive retrieval accuracy and computational efficiency, and it also showed strong performance across the conditions examined in this study. The comparison therefore focuses on FAST-based discrete candidate matching and AecroFormer-based continuous regression under the same Lorenz–Mie forward model, $3\beta + 2\alpha$ optical constraints, and parameter domain. The sensitivity of FAST to the retained candidate number and aggregation weight is further examined in Supplementary Section F, where moderate parameter changes do not alter the main comparative conclusions. Regularization-based inversion, TiARA, and maximum-likelihood retrieval involve different prior, regularization, and optimization assumptions and are therefore outside the scope of the present controlled comparison.

To avoid interference from fixed sampling structures, three independently generated non-gridded test sets were constructed by random sampling in the continuous parameter space followed by Lorenz–Mie forward simulation. Each test set contained 800,000 samples and was fully independent of the training data. None of the test sets was used for model training, early stopping, hyperparameter selection, or uncertainty calibration. Each deep-learning configuration was independently trained using three predefined random seeds, and each trained model was evaluated on all three test sets under the same standard-noise specification. Consequently, nine evaluation results were obtained for each deep-learning configuration, and their mean and standard deviation are reported in Tables II and III. This protocol provides an out-of-sample assessment of estimator generalization across both training randomness and independent test-set realizations in

continuous parameter space, rather than measuring memorization of a fixed discrete sample library or interpolation among predefined grid points.

2) Standard noise setting

To ensure a objective comparison among different configurations, all statistical evaluations were conducted under a unified standard noise condition. This setting corresponds to the relative Gaussian noise standard deviations of the optical inputs obtained from a 3β+2α lidar observation system, with noise injected independently into each channel. Using the notation introduced earlier, let the basic optical observation vector be $\boldsymbol{y}$, and let the noisy observation vector be denoted by $\tilde{\boldsymbol{y}}$. Then

$$\tilde{y}_k = y_k(1+\epsilon_k), \epsilon_k \sim \mathcal{N}(0, \tau_k^2), k = 1, \dots, 5. \quad (21)$$

Following the test setting used in previous work, the relative noise standard deviations for the five channels were fixed as

$$(\tau_{\alpha_{355}}, \tau_{\beta_{355}}, \tau_{\alpha_{532}}, \tau_{\beta_{532}}, \tau_{\beta_{1064}}) = (0.10, 0.05, 0.10, 0.05, 0.15).$$

TABLE I. COMPOSITION OF THE COMPARED ABLATION CONFIGURATIONS

Configuration	Reorganized input representation	Physical consistency constraint	Dynamic non-gridded sampling	Heteroscedastic uncertainty modeling	Data augmentation	Description
ORI	×	×	×	×	√	Original AecroFormer reference
ORI-P	×	√	×	×	√	ORI with the physical consistency constraint
LUT	√	×	×	×	×	KNN-based lookup table retrieval (FAST)
BAS	√	×	×	×	√	Unified baseline proposed in this study
BAS-P	√	√	×	×	√	BAS with the physical consistency constraint
BAS-D	√	×	√	×	√	BAS with dynamic non-gridded sampling
DYN	√	√	√	×	√	BAS-P with dynamic non-gridded sampling
UNC	√	√	√	√	√	DYN with heteroscedastic uncertainty modeling
NOA	√	√	√	√	×	UNC without data augmentation

That is, the extinction coefficients at 355 nm and 532 nm were assigned 10% noise, the backscatter coefficients at 355 nm and 532 nm were assigned 5% noise, and the backscatter coefficient at 1064 nm was assigned 15% noise. This setting defines the standard noise level used throughout this paper and serves as the common evaluation protocol for all subsequent tables and figures.

Under this setting, the output of any method can be uniformly denoted as $\hat{\theta}(\tilde{y})$, that is, the point estimate obtained from the noisy observation vector under the standard noise perturbation. For the deep learning configurations used here, the actual network input is the feature vector $\tilde{\boldsymbol{z}}$ further constructed from $\tilde{y}$. However, for simplicity of notation, this section still uses $\hat{\theta}(\tilde{y})$ to represent the retrieval result driven by noisy observations. All subsequent error statistics are defined on this common notation.

3) Evaluation metrics

To comprehensively evaluate the effects of different strategies on the retrieval results, four metrics were adopted for different variables: mean absolute error (MAE), mean error (ME), mean absolute relative error (MARE), and mean relative error (MRE). For the real part m_r and the imaginary part m_i of the complex refractive index, absolute error metrics were

mainly used. For variables related to particle size, distribution width, and particle number concentration, relative error metrics were used because they are more appropriate on a relative scale. In the result tables, in addition to the variables directly predicted by the network, metrics are also reported for post-retrieval quantities. The aerosol volume concentration is recovered from the absolute optical magnitude after retrieval of the normalized microphysical state, and the particle number concentration is subsequently derived from the recovered volume concentration and particle size-distribution parameters.

Let the true and predicted values of the jth target variable for the ith sample be denoted by $\theta_{j,i}$ and $\widehat{\theta_{j,i}}$, respectively, and let the total number of samples be n. The four metrics are defined as follows:

$$\mathrm{MAE}_j = \frac{1}{n}\sum_{i=1}^{n} | \hat{\theta}_{ij} - \theta_{ij} |, \tag{22}$$

$$\mathrm{ME}_j = \frac{1}{n}\sum_{i=1}^{n} \left(\hat{\theta}_{ij} - \theta_{ij}\right), \tag{23}$$

$$\mathrm{MARE}_j = \frac{1}{n}\sum_{i=1}^{n} \left| \frac{\hat{\theta}_{ij} - \theta_{ij}}{\theta_{ij}} \right| \times 100\%, \tag{24}$$

$$\mathrm{MRE}_j = \frac{1}{n}\sum_{i=1}^{n} \frac{\hat{\theta}_{ij} - \theta_{ij}}{\theta_{ij}} \times 100\%. \tag{25}$$

Here, MAE and MARE describe the average magnitude of the error and are used to measure retrieval accuracy, whereas ME and MRE preserve the sign of the error and are used to characterize whether the estimator tends to systematically overestimate or underestimate the target. In other words, the absolute error metrics answer the question “how far is the estimate from the truth,” whereas the signed error metrics answer the question “in which direction does the estimator tend to deviate.” In Section III , these metrics are mainly used to report the performance differences under the unified evaluation protocol, whereas their evolution with noise level and perturbation conditions is further analyzed in Section IV.

4) Ablation results and analysis

TABLE II. ERROR MAGNITUDE UNDER THE STANDARD NOISE SETTING FOR DIFFERENT ABLATION CONFIGURATIONS

Method	m_{r}**MAE**	m_{i} **MAE**	r_{med} **MARE (%)**	σ_{g} **MARE (%)**	N **MARE (%)**
ORI	0.0615 ± 0.0001	0.0198 ± 0.0001	15.49 ± 0.04	2.901 ± 0.004	67.33 ± 2.44
ORI-P	0.0616 ± 0.0004	0.0210 ± 0.0002	16.11 ± 0.04	2.897 ± 0.004	132.51 ± 24.24
LUT	0.0673 ± 0.0004	0.0212 ± 0.0001	14.21 ± 0.03	3.998 ± 0.008	86.65 ± 1.29
BAS	0.0496 ± 0.0002	**0.0172 ± 0.0001**	13.08 ± 0.09	2.860 ± 0.004	44.70 ± 0.62
BAS-P	0.0494 ± 0.0001	0.0178 ± 0.0001	13.49 ± 0.04	2.855 ± 0.004	60.45 ± 2.67
BAS-D	0.0496 ± 0.0002	0.0173 ± 0.0001	13.03 ± 0.06	2.854 ± 0.005	**44.13 ± 1.30**
DYN	**0.0492 ± 0.0002**	0.0175 ± 0.0001	13.42 ± 0.07	2.853 ± 0.004	57.63 ± 2.30
UNC	0.0494 ± 0.0001	0.0174 ± 0.0001	**12.93 ± 0.05**	**2.853 ± 0.003**	51.58 ± 0.91
NOA	0.0803 ± 0.0020	0.0252 ± 0.0004	16.96 ± 0.52	4.104 ± 0.056	113.65 ± 27.28

TABLE III. MEAN SIGNED ERROR UNDER THE STANDARD NOISE SETTING FOR DIFFERENT ABLATION CONFIGURATIONS

Method	m_{r} **Abs. ME**	m_{i} **Abs. ME**	r_{med} **Abs. MRE** (%)	σ_{g} **Abs. MRE** (%)	N **Abs. MRE** (%)
ORI	0.0030 ± 0.0021	0.0016 ± 0.0006	3.52 ± 0.42	0.064 ± 0.021	30.23 ± 3.03
ORI-P	0.0060 ± 0.0008	0.0011 ± 0.0004	2.50 ± 0.71	0.064 ± 0.036	101.88 ± 25.35
LUT	0.0163 ± 0.0002	0.0031 ± 0.0000	1.53 ± 0.06	0.547 ± 0.011	56.66 ± 1.40
BAS	0.0011 ± 0.0009	0.0006 ± 0.0003	2.53 ± 0.39	0.077 ± 0.019	**14.05 ± 0.67**
BAS-P	0.0042 ± 0.0006	0.0008 ± 0.0004	**0.23 ± 0.14**	0.043 ± 0.015	36.68 ± 3.23
BAS-D	**0.0006 ± 0.0006**	0.0004 ± 0.0002	1.94 ± 0.65	**0.023 ± 0.015**	14.52 ± 2.20
DYN	0.0048 ± 0.0009	0.0005 ± 0.0004	0.46 ± 0.42	0.072 ± 0.022	33.41 ± 2.59
UNC	0.0051 ± 0.0005	**0.0003 ± 0.0001**	1.06 ± 0.18	0.065 ± 0.010	26.25 ± 1.22
NOA	0.0260 ± 0.0044	0.0026 ± 0.0017	3.33 ± 1.06	0.900 ± 0.182	83.13 ± 29.68

Tables II and III summarize the error magnitudes and mean signed errors of the compared configurations under the standard-noise setting. To assess sensitivity to training and test randomness, each deep-learning configuration was independently trained using three predefined random seeds. Each trained model was subsequently evaluated on three independently generated test sets following the same standard-noise specification. The reported mean and standard deviation were therefore calculated from nine evaluations for each deep-learning configuration. The standard deviation reflects the combined variability associated with model training and independent test-set realizations.

The results reveal that the proposed components play different and complementary roles rather than producing a uniformly monotonic improvement in every metric. The most consistent gain in point-estimate accuracy is obtained from the reorganized input representation. Compared with ORI, BAS reduces the MAE of $m_{\mathrm{r}}$from 0.0615 to 0.0496, corresponding to a reduction of approximately 19.3%, and decreases the MAE of m_{i} from 0.0198 to 0.0172. The MARE of r_{med} decreases from 15.49% to 13.08%, while that of N decreases from 67.33% to 44.70%. By contrast, the change in σ_{g} is relatively small. These results indicate that the reorganized optical representation establishes a more effective baseline for learning the inverse mapping, particularly for the refractive-index, particle-size, and concentration-related quantities.

The physical consistency constraint exhibits a more parameter-dependent effect. Relative to BAS, BAS-P maintains a similar m_{r} error but does not reduce the absolute error for all variables; the errors of m_{i}, r_{med}, and particularly N increase to different degrees. A similar tendency is observed when the physical term is introduced under the original input setting in ORI-P. Nevertheless, Table III shows that the mean signed bias of r_{med} for BAS-P is centered close to zero, decreasing from approximately +2.53% for BAS to +0.11%±0.26%. The physical term should therefore not be interpreted as a universally accuracy-improving component. Instead, it changes the balance among coupled parameters by imposing an additional observation-space closure constraint, which can reduce selected systematic biases while introducing trade-offs in other parameter-space errors. Its direct effect on observation-space consistency is examined separately in the supplementary analysis.

Dynamic non-gridded sampling has a relatively modest independent effect on the average point-estimate errors. BAS-D remains close to BAS across most metrics, indicating that continuously introducing off-grid samples does not by itself produce a large reduction in mean error under the present test condition. When added to BAS-P, however, dynamic sampling partially compensates for the parameter-space error increases associated with the physical term. Compared with BAS-P, DYN reduces the errors of m_{i}, r_{med}, and N, although the resulting values do not uniformly surpass those of BAS or BAS-D. Its contribution is therefore primarily associated with reducing dependence on a fixed discrete training distribution and extending the coverage of the continuous parameter space, rather than with substantially improving every average accuracy metric.

UNC further introduces heteroscedastic uncertainty modeling on top of DYN. Compared with DYN, the MARE of r_{med} decreases from 13.42% to 12.93%, and the MARE of N decreases from 57.63% to 51.58%, while the errors of m_{r}, m_{i}, and σ_{g} remain at comparable levels. UNC is therefore not the uniformly best point estimator for every variable. Its main advantage is that it maintains competitive point-estimate accuracy while additionally providing sample- and parameter-dependent uncertainty estimates. The usefulness of these uncertainty outputs is further evaluated through empirical interval coverage and profile-based field analyses.

The NOA configuration provides the clearest

counterexample. Once augmentation is removed from UNC, the MAE values of m_{r} and m_{i} increase to 0.0803±0.0020 and 0.0252±0.0004, respectively, while the MARE values of r_{med}, σ_{g}, and N increase to 16.96%±0.52%, 4.104%±0.056%, and 113.65%±27.28%. The substantially larger mean errors and inter-run variability indicate that augmentation is not merely an optional optimization technique. Rather, it is necessary for representing the observational perturbations encountered during evaluation and for reducing the distribution shift between idealized forward simulations and noisy optical inputs. This conclusion is limited to the perturbation types and ranges represented during training and should not be interpreted as robustness against arbitrary observational errors.

Taken together, the ablation results reveal a clear division of roles among the proposed components. The reorganized input representation provides the dominant and most consistent improvement in point-estimate accuracy. The physical consistency loss modifies the observation-parameter trade-off and reduces selected signed biases, but does not uniformly improve all microphysical errors. Dynamic non-gridded sampling reduces dependence on a fixed discrete parameter grid and partially mitigates the parameter-space cost introduced by the physical term. Heteroscedastic uncertainty modeling maintains competitive accuracy while adding explicit credibility information, whereas physically guided augmentation is essential for obtaining a stable estimator under perturbed observations. Whether these module-specific effects remain evident under real observational conditions is further examined in Section III.B using the DISCOVER-AQ California dataset.

Because the main incremental role of UNC lies partly in its uncertainty output rather than uniformly superior point-estimate accuracy, the reliability of the predicted intervals is examined separately below.

5) Uncertainty characterization

In addition to point estimates, UNC provides sample-dependent predictive intervals for each directly retrieved microphysical parameter. Their empirical coverage was evaluated on the held-out synthetic test set under the standard-noise setting using the nominal 68% and 95% intervals. The results indicate that the uncertainty estimates are informative but clearly parameter dependent. For m_{r}, the empirical coverage rates are 72.3% and 93.9%, respectively, both reasonably close to their nominal levels, indicating the most consistent interval behavior among the retrieved parameters. For m_{i}, the 95% coverage reaches 94.9%, whereas the 68% coverage increases to 83.9%, suggesting that the narrower interval is relatively conservative. The coverage rates for r_{med} are 65.6% and 91.6%; thus, its uncertainty estimates reasonably characterize the main error range, although some larger deviations remain insufficiently covered by the nominal 95% interval. By contrast, $\ln\sigma_{\mathrm{g}}$shows less consistent interval behavior, with coverage rates of 60.9% and 97.0% for the nominal 68% and 95% intervals, respectively.

These results show that the uncertainty branch is not merely an auxiliary network output. It provides useful credibility information for m_{r}, m_{i}, and r_{med}, although its reliability varies with the retrieved parameter and is more limited for $\ln\sigma_{\mathrm{g}}$. The intervals should therefore be interpreted as parameter-dependent predictive uncertainty under the present training and test distributions, rather than as uniformly calibrated posterior credible intervals. The corresponding uncertainty intervals are further visualized and analyzed in the profile-based field retrievals in Section III.B.2.

6) Retrieval efficiency

Computational efficiency was further assessed for the microphysical inversion step, defined here as the conversion of five-channel lidar optical measurements into aerosol refractive-index and size-distribution parameters. To avoid ambiguity, one retrieval sample refers to one height-resolved 3β+2α optical measurement vector, that is, one range-bin-level input consisting of five optical quantities, rather than a complete lidar profile. A batch of 64 retrieval samples therefore corresponds to 64 independent height-resolved input vectors. The benchmark was conducted on an AMD Ryzen 9 9950X 16-Core Processor. File loading, construction of the lookup table, Mie forward calculations, and number-concentration post-processing were excluded from the timing test, so that only the microphysical inversion step itself was compared. For the same batch of 64 retrieval samples, FAST required 124.42 s, corresponding to 1.94 s per sample. In contrast, the uncertainty-aware AecroFormer configuration used in this study required 1.41 ms per batch, corresponding to 0.022 ms per sample. This represents an acceleration of approximately 8.8×10^4 for the same CPU-based inversion task. This benchmark should therefore be interpreted as a comparison of computational cost for retrieving microphysical parameters from the same five-channel optical inputs, rather than as a statement about a change in observational information content.

B. Real-world validation based on NASA in-situ observations obtained from DISCOVER-AQ California

To further evaluate the applicability of the proposed framework under real observational conditions, external validation was conducted using observations from the NASA DISCOVER-AQ California field campaign [49,50]. This campaign provides a well-coordinated airborne dataset for aerosol microphysical retrieval. The NASA Langley HSRL-2 system onboard the B-200 aircraft provided multiwavelength high-spectral-resolution lidar observations, including the 3β+2α optical input required in this study: backscatter coefficients at 355, 532, and 1064 nm and extinction coefficients at 355 and 532 nm. The P-3B aircraft performed in-situ measurements along multiple spiral profiles, providing independent aerosol microphysical references for evaluating the lidar retrieval results. Sawamura et al. systematically compared HSRL-2 aerosol microphysical retrievals with P-3B in-situ observations during DISCOVER-AQ 2013 California, demonstrating the suitability of this coordinated dataset for evaluating multiwavelength airborne lidar retrievals [51]. Subsequent studies by Wang, Zhou, and co-workers also used DISCOVER-AQ cases to validate and compare FAST and

related lookup-table retrieval methods [28,29]. Therefore, this dataset provides both a realistic field-observation benchmark and an evaluation basis consistent with previous LUT/FAST-based studies.

The matchup procedure used here follows the pairing logic adopted by Sawamura et al. and by Wang and Zhou in their DISCOVER-AQ analyses [28,29,51]. The P-3B spiral was used as the basic matchup unit. When available, spirals were identified using ProfileNumber; otherwise, an auxiliary segmentation based on vertical flight structure was applied. To ensure sufficient vertical representativeness, each spiral was required to last at least 8 min and cover a vertical span of at least 1.5 km. Candidate HSRL-2 segments were searched within a time window from 15 min before the start of the spiral flight-pattern to 15 min after the end of the spiral and were required to be within 15 km of the spiral center. The HSRL-2 and in-situ observations were then resampled to a common resolution of 75 m in altitude and 90 s in time for point-by-point comparison.

Considering that the retrieval framework assumes a unimodal fine-mode particle size distribution, an additional Ångström exponent (AE) constraint was applied to select observations more consistent with this representation. Specifically, $AE_{355-532}$ was restricted to 1.5–2.5, thereby preferentially retaining fine-mode-dominated cases. The $AE_{355-532} = 1.5–2.5$ criterion excluded 67.29% of the valid collocated altitude levels. This relatively high exclusion fraction results from restricting the validation samples to conditions consistent with the fine-mode-dominated unimodal parameterization adopted in this study and therefore limits the generalizability of the field validation. This criterion improves consistency between the field observations and the adopted forward-model assumptions but does not imply that the ambient particle size distributions were strictly unimodal. The 532-nm depolarization ratio was retained as an auxiliary indicator of particle nonsphericity and possible coarse-mode influence, rather than being used as a primary hard-screening criterion.

After profile matching and vertical resampling, a height-resolved sample was retained only when all five HSRL-2 optical inputs and the required in situ reference variables were simultaneously available. Samples containing non-finite or negative values, physically inconsistent spectral relationships, very weak extinction or backscatter signals, or high relative humidity were further excluded. This procedure yielded 31 primary matched profiles from 8 observation days and 14 B-200 flight segments. After point-level quality control, 30 profiles retained valid comparison samples and were used for quantitative evaluation.

Accordingly, the present external validation should be interpreted as an applicability assessment under the selected fine-mode-dominated DISCOVER-AQ California conditions, rather than as evidence of broad generalization across aerosol types, humidity regimes, geographic regions, or particle morphologies.

1) Full-sample validation statistics

TABLE IV. EXTERNAL VALIDATION RESULTS FOR EFFECTIVE RADIUS AND 532-NM SINGLE-SCATTERING ALBEDO USING NASA DISCOVER-AQ CALIFORNIA OBSERVATIONS

Method	r_e MARE (%)	r_e MRE (%)	SSA_{532} MAE	SSA_{532} ME
ORI	25.20 ± 0.35	19.73 ± 0.24	0.0223 ± 0.0009	0.0006 ± 0.0118
ORI-P	23.75 ± 1.41	16.56 ± 1.56	0.0368 ± 0.0044	-0.0296 ± 0.0053
LUT	14.36	-10.92	0.0370	-0.027
BAS	16.10 ± 1.32	1.26 ± 2.07	0.0322 ± 0.0130	0.0076 ± 0.0168
BAS-P	15.21 ± 1.31	-7.64 ± 4.14	0.0288 ± 0.0049	-0.0020 ± 0.0167
BAS-D	14.36 ± 0.86	-4.77 ± 0.70	0.0257 ± 0.0013	0.0142 ± 0.0074
DYN	16.50 ± 2.70	-4.90 ± 3.98	0.0225 ± 0.0032	0.0038 ± 0.0019
UNC	14.59 ± 1.67	-3.99 ± 1.26	0.0247 ± 0.0032	0.0122 ± 0.0054
NOA	29.36 ± 18.98	21.07 ± 20.52	0.1104 ± 0.0747	-0.1004 ± 0.0816

Table IV summarizes the external validation results for the DISCOVER-AQ California matchup dataset. No DISCOVER-AQ observations were used for model training or fine-tuning. All deep-learning configurations were trained exclusively on synthetic data and subsequently evaluated, without retraining, on 139 quality-controlled height-resolved 3β+2α observation vectors retained from 30 matched profiles. For each configuration, three models were independently trained using three predefined random seeds, and the reported values represent the mean and standard deviation across the three trained models. The reported values therefore represent the mean and standard deviation across the three trained models. Here, r_e denotes the effective radius derived from the retrieved particle size distribution, and SSA_{532} denotes the single-scattering albedo at 532 nm. MARE and MAE characterize the error magnitude, whereas MRE and ME retain the direction of the mean deviation relative to the P-3B in situ reference. When the standard deviation of a signed metric spans zero, the corresponding bias direction is not regarded as consistent across training seeds.

Overall, the field results confirm that the proposed modules have distinct and parameter-dependent effects, rather than

forming a strictly progressive sequence in which each additional component improves every metric. The most evident improvement relative to ORI occurs for r_e, whereas the retrieval of SSA_{532} also shows strong overall performance, although its detailed behavior depends more strongly on the specific combination of input representation, physical constraint, sampling strategy, and augmentation.

For r_e, ORI yields a MARE of 25.20%±0.35% and a positive MRE of 19.73%±0.24%, indicating a consistent overestimation of effective radius. Introducing the physical consistency term under the original input setting slightly reduces the MARE to 23.75%±1.41%, but ORI-P retains a substantial positive bias of 16.56%±1.56%. In comparison, BAS reduces the MARE to 16.10%±1.32%, corresponding to an improvement of approximately 36% relative to ORI. Its MRE of 1.26%±2.07% is centered near zero, although the sign varies among training seeds. These results indicate that the reorganized input representation provides the dominant improvement in the external retrieval of particle size, consistent with the results obtained in the synthetic experiments.

The comparisons among the BAS-based configurations further show that the physical consistency loss and dynamic non-gridded sampling affect the r_e retrieval differently. BAS-P reduces the MARE to 15.21%±1.31% but introduces a clearer negative bias of -7.64%±4.14%. BAS-D achieves a MARE of 14.36%±0.86%, equal to the LUT value of 14.36%, while its MRE magnitude is substantially smaller than that of LUT, -4.77%±0.70% compared with -10.92%. When the physical constraint and dynamic sampling are combined, DYN gives a MARE of 16.50%±2.70%, showing that their effects are not strictly additive for particle-size accuracy. UNC subsequently reduces the MARE to 14.59%±1.67% and yields an MRE of -3.99%±1.26%. Thus, UNC remains comparable to LUT in r_e error magnitude while exhibiting a smaller systematic-bias magnitude under the selected field conditions.

The behavior of SSA_{532} is more complex. ORI already achieves a relatively low MAE of 0.0223±0.0009, although its ME of 0.0006±0.0118 shows that the bias direction is not consistent among the three training seeds. This low SSA_{532} error coexists with the much larger r_e error of ORI, illustrating that performance judged from a single derived quantity does not necessarily reflect the overall quality of the microphysical estimator. Adding the physical term directly to the original configuration degrades the absorption-related retrieval: ORI-P increases the MAE to 0.0368±0.0044 and produces a consistent negative ME of -0.0296±0.0053.

BAS alone also does not improve SSA_{532} relative to ORI, yielding an MAE of 0.0322±0.0130, together with relatively large variability among training seeds. However, the additional training mechanisms progressively recover the SSA_{532} performance within the BAS-based framework. BAS-P and BAS-D reduce the MAE to 0.0288±0.0049 and 0.0257±0.0013, respectively. Among the configurations based on the reorganized input representation, DYN achieves the lowest MAE of 0.0225±0.0032, together with a relatively small ME of 0.0038±0.0019. UNC yields an MAE of 0.0247±0.0032, approximately 33% lower than the LUT value of 0.0370, although it retains a positive ME of 0.0122±0.0054. The field results therefore do not support identifying UNC as the uniformly optimal configuration for SSA_{532}; rather, it maintains competitive absorption-related accuracy while additionally providing parameter- and sample-dependent uncertainty information.

The NOA configuration provides the clearest evidence for the importance of data augmentation under field conditions. After augmentation is removed from UNC, the r_e MARE increases to 29.36%±18.98%, approximately twice the UNC value, while the SSA_{532} MAE increases to 0.1104±0.0747, more than four times that of UNC. The markedly enlarged standard deviations indicate that the model also becomes highly sensitive to the training seed. Moreover, NOA exhibits a positive r_e bias of 21.07%±20.52% and a pronounced negative SSA_{532} bias of -0.1004±0.0816. This result supports the interpretation that physically guided augmentation is essential for reducing the discrepancy between idealized synthetic samples and real lidar observations. Nevertheless, this robustness remains bounded by the types and amplitudes of observational perturbations represented during training.

Taken together, the external validation supports the module-specific interpretation obtained from the synthetic experiments while revealing stronger trade-offs under real observations. The reorganized input representation provides the clearest improvement in r_e, but its effect on SSA_{532} depends on the accompanying training mechanisms. The physical consistency loss modifies the bias structure and parameter compensation but does not guarantee lower errors for either derived quantity. Dynamic non-gridded sampling contributes to competitive field performance, particularly when combined with the physical term, whereas augmentation is essential for both accuracy and inter-seed stability. UNC is therefore retained for the subsequent curtain and profile analyses not because it achieves the lowest value for every individual metric, but because it provides a balanced combination of effective-radius accuracy, absorption-related performance, and explicit predictive uncertainty. Its uncertainty outputs are evaluated separately and are subsequently used to support the interpretation of the profile-level retrieval results.

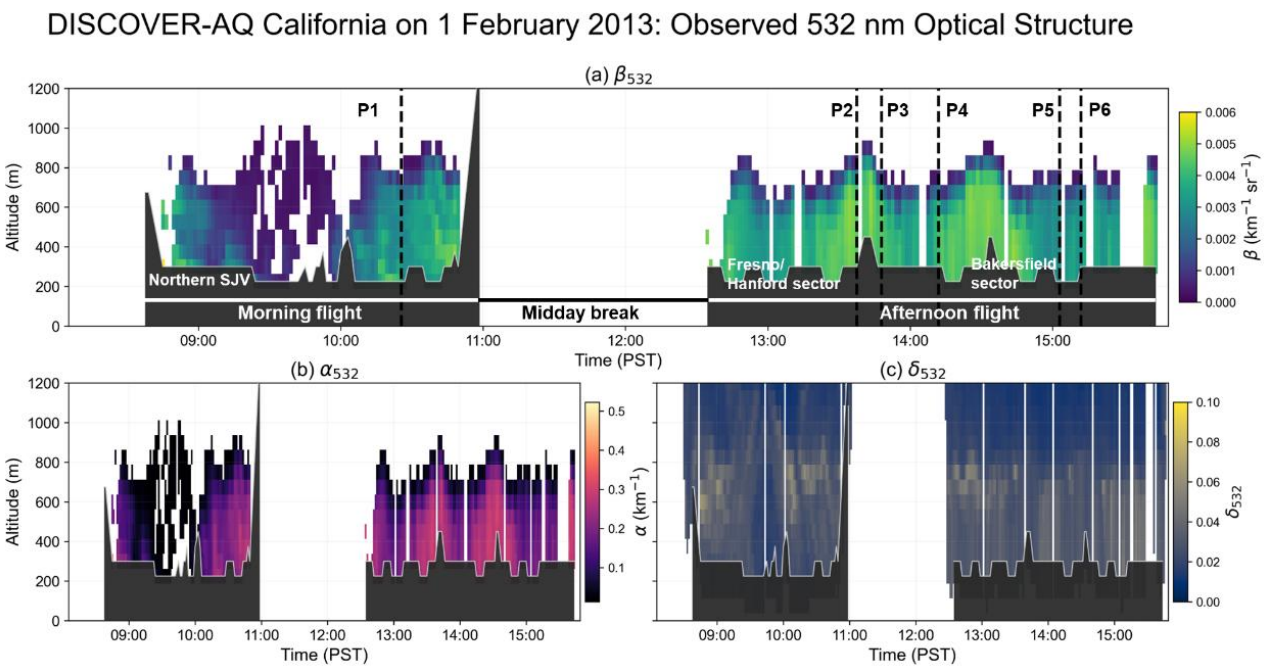


Fig. 2. Observed 532 nm optical structure during the DISCOVER-AQ California flight on 1 February 2013. Panels show (a) the aerosol backscatter coefficient at 532 nm, β_{532}, (b) the aerosol extinction coefficient at 532 nm, α_{532}, and (c) the linear particle depolarization ratio at 532 nm, δ_{532}. The dashed vertical lines P1–P6 indicate the six matched locations used for in-situ profile validation. The morning flight mainly sampled the northern San Joaquin Valley, while the afternoon flight covered the Fresno/Hanford and Bakersfield sectors. Altitude is given relative to mean sea level.

2) Case-based validation using curtain plots and in-situ profiles

To examine how this statistical behavior appears in a realistic two-dimensional observation scene, the DISCOVER-AQ California flight on 1 February 2013 was selected as a representative case for spatiotemporal analysis. This day was selected because it includes both morning and afternoon B-200 science flight segments, covering the regions marked in the figure as Northern SJV, Fresno/Hanford sector, and Bakersfield sector. It also contains a weak elevated aerosol layer, enhanced low-level aerosol loading, regional loading differences, and multiple matched profiles for in-situ validation. According to the NASA HSRL-2 quicklook report for this day, the morning flight used a new flight pattern toward the northern SJV, and the aerosol optical thickness (AOT) showed clear spatial variability during approximately 09:00–11:00 PST (Pacific Standard Time). The afternoon flight followed the standard DISCOVER-AQ pattern. During approximately 13:00–15:30 PST, AOT increased along the flight from the Fresno/Hanford sector toward the Bakersfield sector and reached relatively high values near the Bakersfield-sector sampling period [52]. Therefore, this case is suitable for examining how different retrieval methods respond to realistic regional pollution gradients and vertical structural variability.

Figure 2 shows strong spatiotemporal heterogeneity of the aerosols on 1 February 2013. During the morning flight, a weak elevated aerosol layer appears at approximately 0.4–0.9 km above mean sea level around 09:00–10:00 PST. This layer appears as a weak β_{532} structure in Fig. 2(a), is also weak in α_{532} (Fig. 2(b)), and shows low δ_{532} (Fig. 2(c)). These optical signatures indicate that the layer is not dominated by nonspherical coarse-mode dust or strongly absorbing aerosol but is more consistent with near-spherical fine-mode pollution aerosol, which makes this case an ideal candidate for the comparison/validation test. After 10:00 PST, the low-level aerosol structure becomes more fragmented, with stronger vertical gradients. During the afternoon flight, a more continuous low-level aerosol load is observed over the Fresno/Hanford sector, followed by stronger low-level structures near the Bakersfield sector. This is consistent with the NASA report, which describes enhanced AOT during the afternoon sampling and relatively high AOT near the Bakersfield-related flight segment.

Based on the optical structures shown in Fig. 2, Fig. 3 further compares how LUT (FAST) and UNC interpret the same observation field in terms of aerosol microphysical properties. Both methods capture the main spatiotemporal heterogeneity of the case, including the weak elevated layer in the morning, the enhanced low-level aerosol loading in the afternoon, and the stronger aerosol load near the Bakersfield sector. This result indicates that both methods respond to the major aerosol structures observed by HSRL-2.

However, the two methods decompose the same optical structures into different combinations of microphysical parameters. LUT (FAST) produces locally higher r_e, m_r, and m_i in several regions, and these high values often appear as fragmented patches, as shown in Fig. 3(a), Fig. 3(e), and Fig. 3(g). In contrast, UNC shows smoother changes in loading- and absorption-related variables, with more continuous covariation among r_e, SSA_{532}, m_i, and N, as shown in Fig. 3(b), Fig.

3(d), Fig. 3(h), and Fig. 3(j). These differences should not be interpreted as direct proof that one retrieval is the truth; rather, they need to be assessed jointly with the optical constraints in Fig. 2 and the in-situ profile comparisons at P1–P6.

For the weak (in terms of AOD) elevated layer around 09:00–10:00 PST, Fig. 2 provides a relatively clear external constraint: the layer features weak backscatter and extinction signals and a low depolarization ratio. It is therefore not dominated by nonspherical coarse-mode dust or strongly absorbing aerosol. Under this optical condition, a retrieval that simultaneously produces a large effective radius, high real refractive index, and high imaginary refractive index should be interpreted with caution, because such a parameter combination is not directly supported by strong depolarization, strong extinction, or clear absorption-related optical evidence. In Fig. 3, LUT (FAST) more readily produces such local high-value combinations near this layer, including fragmented high-r_e patches in Fig. 3(a), higher $m_r$values in Fig. 3(e), enhanced $m_i$values in Fig. 3(g), and less continuous variations in N in Fig. 3(i). By contrast, UNC shows a weak but continuous enhancement in N in Fig. 3(j), only moderate adjustment of r_e in Fig. 3(b), no obvious enhancement of m_i in Fig. 3(h), and only small variations in SSA_{532} under a high-SSA background in Fig. 3(d). Therefore, for this weak-layer case, the UNC response is more consistent with the low-depolarization and weak-loading optical constraints, whereas the local high values in LUT (FAST) more likely reflect parameter compensation in an ill-posed retrieval setting.

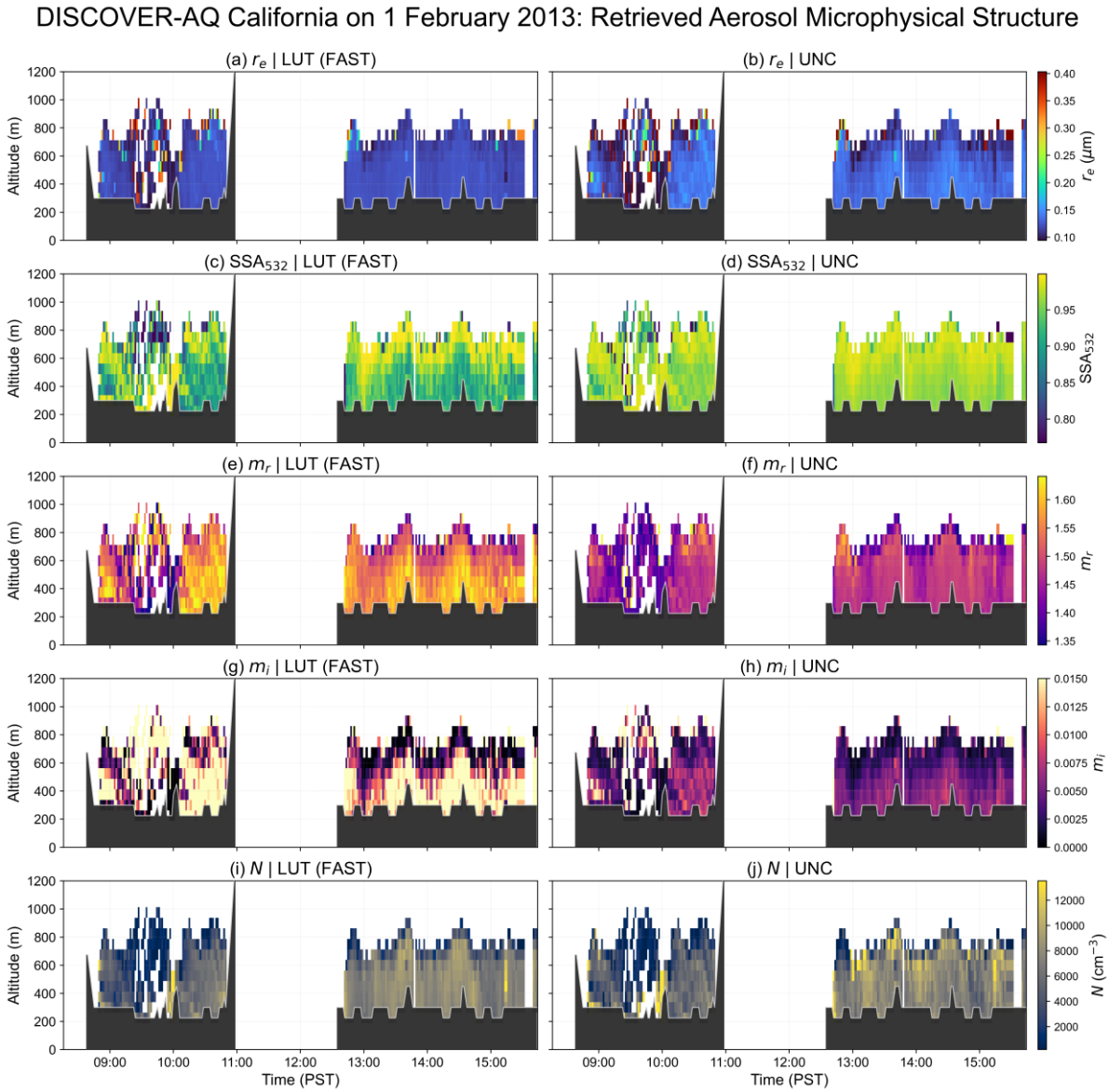


Fig. 3. Retrieved aerosol microphysical structure from LUT (FAST) and UNC on 1 February 2013. Panels show (a, b) effective radius r_e, (c, d) single-scattering albedo at 532 nm SSA_{532}, (e, f) real part of the complex refractive index m_r, (g, h) imaginary part m_i, and (i, j) particle number concentration N. The left column corresponds to LUT (FAST), and the right column corresponds to UNC. The retrievals are based on HSRL-2 optical inputs with a temporal resolution of 90 s and a vertical resolution of 75 m.

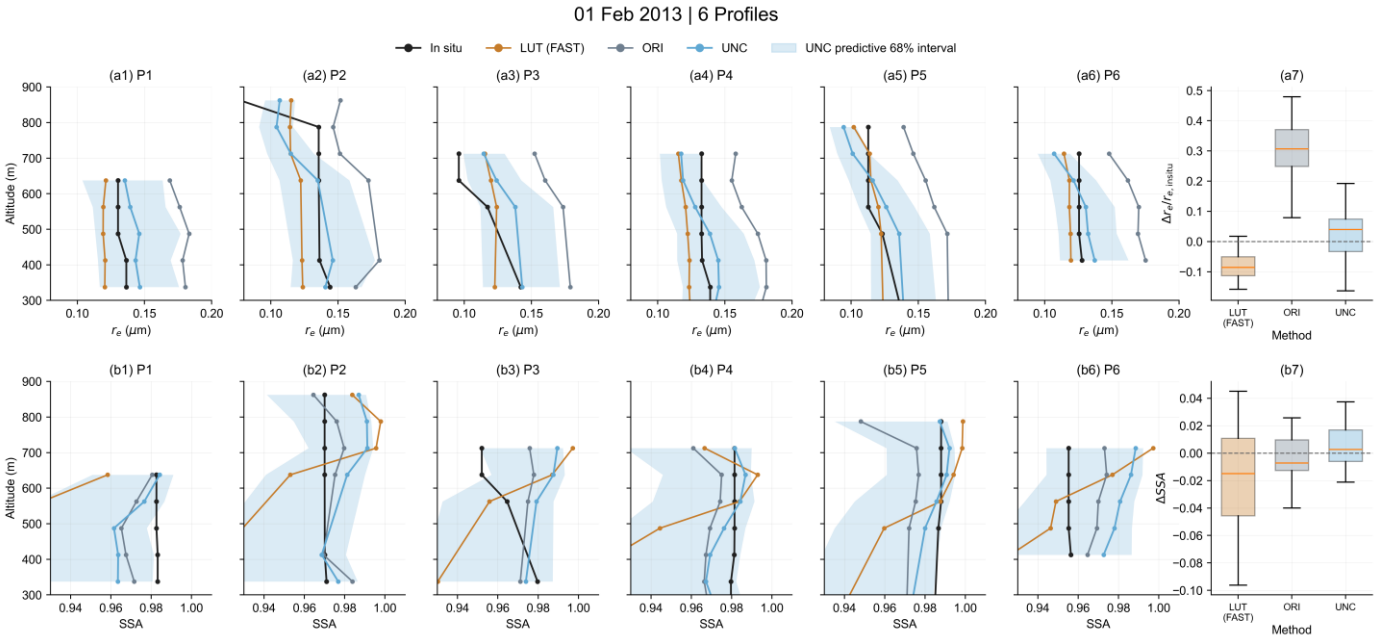


Fig. 4. Profile-level validation at six matched locations on 1 February 2013. P1–P6 correspond to the dashed vertical lines in Figs. 2. Panels (a1)–(a6) show the vertical profiles of effective radius r_e, and panel (a7) summarizes the corresponding signed relative differences, $\Delta r_e/r_{e,\mathrm{in\,situ}}$. Panels (b1)–(b6) show the vertical profiles of SSA_{532}, and panel (b7) summarizes the signed differences. Black, orange, gray, and blue lines denote the P-3B in situ observations, LUT (FAST), ORI, and UNC, respectively. The light-blue shading denotes the central 68% predictive interval provided by UNC for the derived r_e and SSA_{532} profiles. The UNC line represents the ensemble point estimate from the independently trained models. Dashed horizontal lines in panels (a7) and (b7) indicate zero difference. The predictive intervals characterize model-derived retrieval uncertainty and do not include uncertainty associated with spatiotemporal collocation or the in situ reference measurements.

The afternoon segment can be interpreted in the same way. Figures 2(a) and 2(b) show that low-level aerosol loading increases from the Fresno/Hanford sector toward the Bakersfield sector. The NASA report also indicates that AOT increased during the afternoon flight and reached relatively high values near the Bakersfield-sector sampling period. This optical background supports a continuous increase in pollution loading rather than an abrupt change in aerosol type. Correspondingly, UNC shows more continuous variations in N, r_e, m_i, and SSA_{532}, mainly reflecting moderate microphysical adjustments under enhanced aerosol loading. LUT (FAST), however, shows more fragmented high-r_e, high-m_r, or high-m_i patches in some time–height regions. Because these local high values are not always supported by corresponding strong features in β_{532}, α_{532}, or δ_{532}, they are more likely related to jump-like behavior of lookup-table matching in weakly constrained or multi-solution regions. Thus, the 1 February case does not demonstrate that UNC is closer to the in-situ reference at every comparison level. Rather, it shows that, in a complex but continuous real aerosol field, UNC exhibits stronger consistency among the observed optical structure, retrieved parameter covariation, and spatial continuity. Together with the P1–P6 in-situ profile comparisons, these results support the interpretation that UNC provides a more representative retrieval of the aerosol conditions than LUT (FAST) for this case, although the in-situ observations themselves are not treated as an absolute truth.

The smoother UNC curtain should be interpreted as evidence of a more continuous estimator response, rather than direct verification of greater physical accuracy. Although abrupt retrieval changes caused solely by small observational fluctuations are undesirable within a relatively homogeneous aerosol layer, genuine aerosol-layer boundaries and rapid changes in aerosol state may also produce physically meaningful discontinuities. The curtain smoothness is therefore considered together with, rather than as a substitute for, the pointwise in-situ comparisons.

The P1–P6 profile comparisons provide an additional height-resolved reference for evaluating both the point estimates and the predictive uncertainty of UNC. For r_e, ORI shows a systematic positive deviation across most profiles, whereas LUT (FAST) generally produces lower values and a predominantly negative relative difference. UNC is not the closest estimate at every altitude, but its profiles generally remain closer to the in-situ observations than ORI and avoid the more persistent negative deviation of LUT. This behavior is also reflected in panel (a7), where the UNC relative-difference distribution is centered closer to zero and is substantially narrower than that of ORI.

The 68% predictive intervals of UNC frequently overlap the in-situ r_e profiles, while their widths vary with altitude and profile. Wider intervals occur at several levels where the point estimates of the different methods also diverge more strongly, indicating that the uncertainty branch identifies some of the

locally weakly constrained retrieval conditions. Nevertheless, interval overlap should not be interpreted as pointwise verification of physical correctness, because the predictive intervals describe uncertainty in the learned estimator and do not include errors in the collocation or in-situ reference.

For SSA_{532}, the differences among the three retrieval methods are generally smaller in absolute magnitude than those for r_e, but their vertical behavior remains parameter and profile dependent. LUT (FAST) exhibits a broader and predominantly negative difference distribution, whereas the signed differences of ORI and UNC are more concentrated around zero for this case. UNC does not uniformly outperform ORI at every altitude, but it additionally provides predictive intervals that indicate substantial variation in absorption-related retrieval confidence. In several weakly constrained altitude ranges, the SSA_{532} intervals are relatively broad, consistent with the parameter-dependent uncertainty behavior identified in the analysis of the synthetic data. Thus, the value of UNC in Fig. 4 lies not only in its point estimate, but also in explicitly indicating where the retrieved profile should be interpreted with greater caution.

It should be noted that discrepancies between the retrievals and the P-3B in-situ observations remain, even for UNC. These differences should not be attributed solely to retrieval errors. On the one hand, aerosol microphysical retrieval from multiwavelength lidar is an ill-posed inverse problem, and the limited 3β+2α optical constraints cannot uniquely determine all microphysical parameters. On the other hand, the field comparison is also affected by imperfect temporal and spatial collocation between the B-200 and P-3B aircraft, horizontal drift, humidity correction of the in-situ observations, the representativeness of in-situ sampling, and strong vertical gradients. In addition, the two observing systems involve different averaging characteristics: HSRL-2 represents an optical-volume average within each time–height bin, whereas the P-3B in-situ reference is obtained from measurements sampled along the aircraft flight path and then averaged within the corresponding altitude interval. In shallow pollution layers or near the top of the boundary layer, these differences in sampling geometry and averaging scale may cause HSRL-2 and P-3B to represent different aerosol volumes even when the matchup criteria are satisfied. Therefore, the results in this section should be interpreted as an assessment of overall external consistency under field conditions, rather than as a point-by-point validation against an error-free reference.

Overall, the DISCOVER-AQ California validation shows that the proposed framework remains applicable beyond controlled synthetic experiments, although its advantages are clearly parameter dependent. The full-sample results demonstrate a substantial improvement in r_e relative to ORI, with BAS-D and UNC reaching error levels comparable to LUT (FAST) while showing smaller signed-bias magnitudes. For SSA_{532}, the augmented BAS-based configurations generally outperform LUT, but they do not uniformly improve upon ORI, indicating a stronger dependence on the training configuration and parameter-compensation structure.

The 1 February 2013 case further demonstrates the practical role of the uncertainty-enabled estimator. UNC provides competitive point estimates and more continuous vertical responses while explicitly identifying altitude ranges with larger predictive spread. The broad intervals at some height levels also show that the uncertainty output should not be interpreted as uniformly precise. Instead, the combination of the point estimate, predictive interval, optical structure, and in-situ comparison provides a more complete basis for interpreting retrieval credibility under field conditions. Thus, UNC is retained for the subsequent analyses because it offers a balanced combination of point-estimate performance, local continuity, and uncertainty-aware interpretation, rather than because it is uniformly optimal for every parameter and altitude.

The external validation primarily demonstrates performance and uncertainty behavior under selected field conditions. It does not, by itself, explain why continuous deep-learning estimators respond differently from discrete LUT matching under ill-posed conditions. Section IV therefore turns to an estimator-level mechanistic analysis based on statistical degradation, local conditional distributions, and perturbation-plane responses.

IV. Perturbation and Noise Analyses: Advantages, Boundaries, and Limitations of Deep Learning in Ill-Posed Retrieval Problems

A central incremental contribution of the present study is to move beyond the performance-oriented comparison adopted in the original AecroFormer work and examine the behavior of the retrieval methods at the estimator level. The purpose of this section is not to argue that deep learning removes the intrinsic ill-posedness of the inverse problem, but to investigate how learned continuous estimation and discrete LUT-based candidate matching respond differently under the same limited optical information and observational perturbations.

Although the results in Section 3 show that the deep learning configurations generally outperform the traditional lookup-table method in overall accuracy, this advantage does not imply any increase in the amount of observational information itself. For the aerosol microphysical retrieval problem from multiwavelength lidar considered here, the inverse problem remains strongly nonlinear, highly coupled, and locally non-identifiable. As a result, small perturbations in observation space can still be substantially amplified in parameter space. Therefore, the essential difference between the different methods does not lie in whether they obtain “more information,” but in how they construct parameter estimates under the same observational uncertainty.

From this perspective, both LUT and deep learning can be regarded as different implementations of estimators constructed from observations. The former is closer to matching and switching over a discrete candidate set [25], whereas the latter is closer to a statistical compromise over a continuous conditional distribution. Consequently, when noise increases and local uncertainty broadens, the two types of estimators face the same observational constraints, but may exhibit markedly different degradation trajectories, bias evolution patterns, and local responses to perturbations. Following this unified

perspective, this section analyzes the source of the advantage of deep learning in ill-posed retrieval, as well as its scope of applicability and limitations, from three levels: statistical degradation, conditional-distribution-based mechanism, and local geometric response to perturbations.

A. Macroscopic statistical degradation trajectories

To examine how different methods degrade as noise increases, we first compare the behavior of the LUT and UNC estimators under different error conditions from a statistical perspective. Here, the noise scaling factor uniformly multiplies the standard channel-specific noise levels defined in Section III. A factor of 0 denotes clean optical inputs without additional noise, a factor of 1 corresponds to the standard-noise setting (10% for the extinction channels, 5% for the 355/532-nm backscatter channels, and 15% for the 1064-nm backscatter channel), and a factor of 2 therefore represents twice these error levels.

Following the evaluation strategy in Section III, mean error (ME) and mean relative error (MRE) are used here to characterize the systematic bias of the estimators and its evolution with noise level; the corresponding definitions are given in Eqs. (23) and (25). Unlike Section III, which focuses on comparisons under a unified evaluation protocol, the purpose here is to determine whether the bias trajectory remains smooth as noise increases, whether bias reversal occurs, and whether the uncertainty band expands rapidly.

TABLE V. STATISTICAL DEGRADATION UNDER INCREASING NOISE

Condition	Method	m_r ME	m_i ME	r_e MRE (%)	N MRE (%)	σ MRE (%)
clean	LUT	-0.003	-0.0002	0.678	-1.115	0.077
clean	UNC	-0.007	-0.0004	-0.298	7.267	0.172
noisy	LUT	0.016	0.0029	-1.035	52.233	-0.605
noisy	UNC	**-0.004**	**0.0003**	0.726	**30.460**	**0.090**

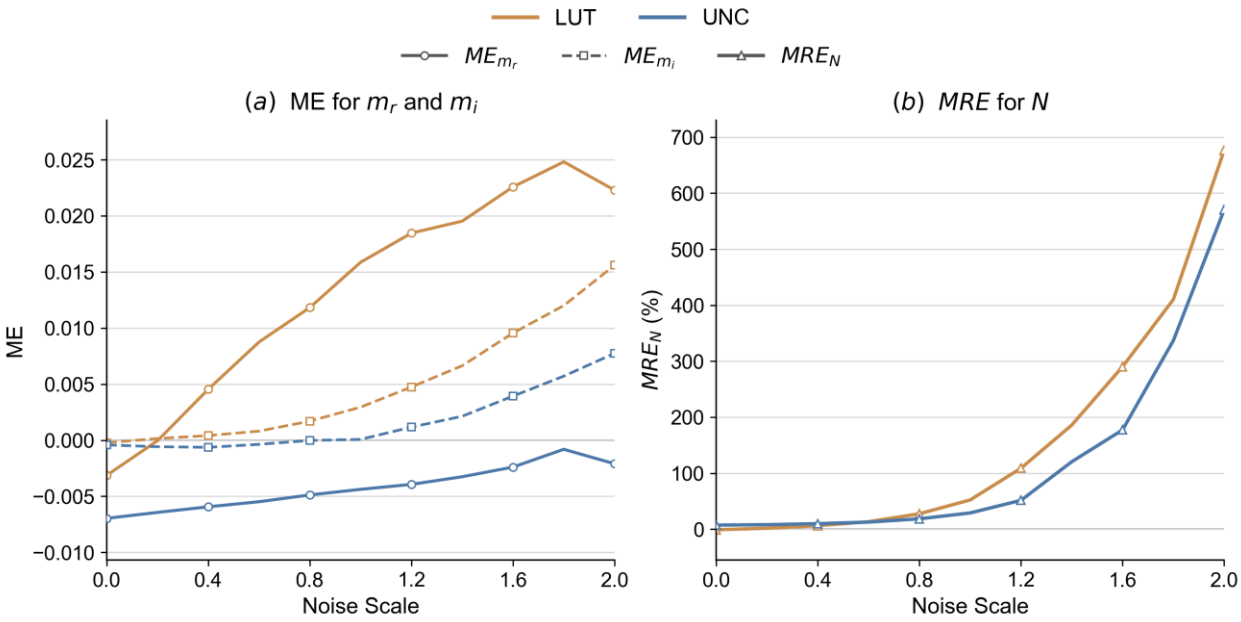


Fig. 5. Statistical degradation curves of LUT and UNC under increasing noise intensity. In panel (a), the mean errors (ME) of m_r and m_i are shown as functions of the noise scaling factor; solid lines with circle markers denote m_r, and dashed lines with square markers denote m_i. In panel (b), the mean relative error (MRE) of particle number concentration N is shown as a function of the noise scaling factor. Orange denotes LUT and blue denotes UNC. This figure is used to illustrate the bias evolution trajectories and degradation differences of the two estimators as noise increases continuously.

The focus of this subsection is not on the performance at any single noise level, but on the overall degradation pattern of different retrieval methods as the noise level increases. Table V summarizes the corresponding clean- and noisy-condition statistics, which are discussed below. Table V gives the statistics of the refractive index, effective radius, particle number concentration, and distribution width parameter under two representative conditions, namely clean and noisy, while Fig. 5 further shows how the bias trajectories of the key parameters evolve as the noise level increases continuously. For clarity, Fig. 5 visualizes selected parameters representing the more pronounced degradation modes. The refractive-index parameters illustrate noise-induced bias drift, whereas N represents strong ill-posed error amplification. The r_e and σ curves are not additionally shown because their variations are substantially smaller and do not introduce a distinct degradation

pattern beyond those already visualized.

Overall, the advantage of deep learning is not expressed uniformly across all parameters but depends strongly on the ill-posedness associated with each variable. Different parameters have different sensitivities to noise and therefore exhibit different degradation modes. In addition, the two methods differ in how they form point estimates under the same observational uncertainty, which further leads to clearly different bias trajectories.

For m_r, LUT has a mean bias of -0.003 under the clean condition, which is slightly better than -0.007 for UNC. Under the noisy condition, however, the mean bias of LUT changes sign and increases to +0.016, whereas UNC changes only from -0.007 to -0.004 and remains within a relatively small negative-bias range. This indicates that m_r is better constrained by the optical observations than strongly ill-posed quantities such as N, but it can still be affected by noise-induced bias drift. The main advantage of UNC is that its bias trajectory is much flatter and remains in a small negative range, whereas the LUT trajectory increases more steeply with noise and changes from negative to positive bias.

The degradation pattern of m_i differs from that of m_r. The bias magnitude remains small under clean and moderate-noise conditions, and a clear increase appears only at higher noise levels. In Fig. 5(a), the m_i trajectories are nearly flat at low noise scales and become steeper only when the perturbation is large. The mean bias of LUT changes from -0.0002 to +0.0029, whereas UNC changes from -0.0004 to +0.0003. This indicates that the degradation of m_i is delayed relative to m_r, and that UNC still provides a smoother bias trajectory under increasing noise.

Compared with the refractive index parameters, N exhibits a more typical strongly ill-posed degradation pattern. The relative error of LUT increases from -1.12% under the clean condition to 52.23% under the noisy condition, whereas the corresponding value for UNC increases from 7.27% to 30.46%. Although both methods deteriorate substantially as noise increases, the growth rate is clearly smaller for UNC. Under the noisy condition, the relative error of UNC for N is about 58.3% of that of LUT. This indicates that, for a strongly ill-posed parameter such as N, deep learning does not remove the ill-posedness itself but significantly delays the onset of instability. In other words, its advantage is mainly reflected in suppressing error explosion rather than recovering a uniquely correct value.

The effective radius r_e and the width parameter σ exhibit another type of degradation behavior that is more characteristic of statistical rebalancing. For r_e, the MRE of LUT changes from 0.68% under the clean condition to -1.03% under the noisy condition, whereas UNC changes from -0.30% to 0.73%. Both estimators undergo a sign change as noise increases, but the bias magnitude remains relatively small overall. For σ, the MRE of LUT changes from 0.08% to -0.61%, whereas UNC changes from 0.17% to 0.09%, showing only a small fluctuation. This suggests that, for size-distribution-related quantities that are influenced jointly by several microphysical parameters, noise-induced changes are more strongly reflected in a rebalancing of the estimate location than in a simple monotonic deterioration.

It should also be noted that not all parameters necessarily follow the simple rule that "larger noise leads to larger error." For example, in Table V, the MRE of LUT for r_e changes from 0.68% under the clean condition to -1.03% under the noisy condition. Such an apparent change more likely reflects a situation in which the estimate is pushed toward a more central statistical location by noise, rather than that noise itself provides additional information. Therefore, for parameters of this kind, conclusions should not be drawn from a few isolated error values alone. Instead, continuous noise curves, median trends, and expansion of dispersion should be considered together.

Taken together, Table V and Fig. 5 show that different parameters exhibit clearly different degradation modes [19,20]. The behavior of m_r is mainly characterized by systematic drift under increasing noise, m_i is more strongly characterized by smooth statistical compromise, and N shows a typical strongly ill-posed amplification process. By contrast, the variation of r_e and σ is closer to local statistical redistribution. Correspondingly, LUT is more likely to exhibit sign reversal, local jumps, and rapid uncertainty expansion as noise increases, whereas UNC generally shows smoother bias trajectories, slower error growth, and milder uncertainty expansion. This indicates that the advantage of deep learning is not a uniform "better in every respect," but rather depends closely on the local identifiability of each parameter and is mainly expressed through smoothing and delaying the degradation process.

B. Estimation mechanism under local conditional distributions

Here, the conditional distribution is considered from the inverse-retrieval perspective, namely $\boldsymbol{p}(\boldsymbol{\theta} \mid \mathbf{y})$: for a given optical observation vector $\mathbf{y}$ within its uncertainty range, multiple microphysical parameter vectors may remain physically compatible with the observation. This differs from the forward measurement distribution $\boldsymbol{p}(\mathbf{y} \mid \boldsymbol{\theta})$, which describes possible optical observations for a fixed microphysical state. The two are related through the forward model and the assumed error structure, but they are not interchangeable in an ill-posed inverse problem.

This empirical conditional-distribution view can be regarded as a statistical counterpart of the low-misfit solution valleys often discussed in deterministic inverse problems [24,53,54]. It describes how multiple microphysical parameter states remain compatible with the same optical observation within the uncertainty bounds.

The previous subsection shows that, under the same increase in noise, different parameters exhibit different degradation patterns, and the same parameter can also show markedly different bias trajectories under different estimators. This means that the difference between the two methods lies not only in the error magnitude, but also in the mechanism by which they produce point estimates under the same observational uncertainty. To further explain this difference, we now turn to the conditional distribution perspective and examine the concentration region of possible parameter values under a given observation, as well as the relative positions of different estimators within that distribution.

Following the unified notation used above, the essence of

retrieval for a given observation y is to construct an estimator $\hat{\theta}(y)$. When noise is small, the local conditional distribution is usually concentrated. As noise increases, however, the same observation neighborhood may correspond to a larger set of candidate parameter states, so that the conditional distribution broadens and may further exhibit skewness or even local multimodality. Based on this idea, Fig. 6 does not directly display the statistical distribution of the full sample set. Instead, it centers on a selected reference observation y_0 and its perturbed observations, uses a kernel-weighted screening of neighboring samples in the sample library according to their distance in observation space, and then constructs one-dimensional empirical conditional densities for each parameter. Therefore, the figure does not show a strict Bayesian posterior distribution, but rather an empirical approximation of the parameter conditional distribution under local observational constraints. Its purpose is to diagnose local identifiability and to compare the positions of the true value, the LUT estimate, and the UNC estimate with respect to the high-density part of the local empirical conditional density. Here, the "high-density part" refers to the region around the dominant mode of the kernel-weighted empirical density, rather than to a formally defined Bayesian credible interval. The relevant kernel bandwidth, neighborhood scale, effective sample size, and sensitivity settings are given in Appendix B.

Within this empirical conditional distribution framework, the mathematical mechanisms of the two methods can be stated more clearly. For LUT, the essence is to search for the candidate in the discrete parameter library Θ_{LUT} that best matches the current observation:

$$\widehat{\boldsymbol{\theta}}_{\mathrm{LUT}}(\mathbf{y}) = \arg \min_{\boldsymbol{\theta}\in\Theta_{\mathrm{LUT}}} \| f(\boldsymbol{\theta}) - \mathbf{y} \|_2^2. \tag{26}$$

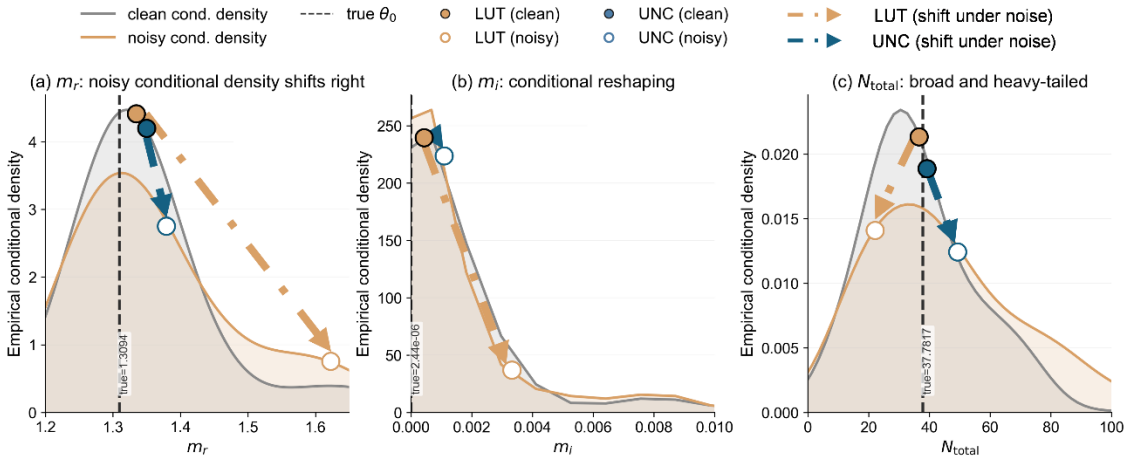


Fig. 6. Local empirical conditional densities of m_r, m_i, and N under clean and noisy conditions. Panels (a)–(c) correspond to m_r, m_i, and N, respectively. The gray and orange curves denote the one-dimensional empirical conditional densities constructed from the local observation neighborhood under the clean and noisy conditions, and the black dashed line denotes the true value θ_0. Filled and hollow circles indicate the LUT and UNC point estimates under the clean and noisy conditions, respectively. Arrows denote the migration direction of the estimates as noise increases. The figure is used to illustrate how different estimators change their relative positions with respect to the main probability mass region as the local conditional distribution broadens. The UNC (clean) and LUT (clean) markers overlap in panel (b).

Therefore, the output of LUT is influenced by the partition of the finite discrete candidate set. When the observation remains within the region associated with the same candidate, the retrieved parameter may stay nearly unchanged. Once the observation crosses the boundary between neighboring candidate regions, however, the selected grid point may change abruptly. This piecewise behavior can in principle be reduced by using a finer LUT resolution. In practice, however, the dimensionality of the aerosol microphysical parameter space makes exhaustive grid refinement computationally expensive and storage-demanding, and potentially intractable when additional degrees of freedom, such as wavelength-dependent refractive index or departures from an ideal lognormal size distribution, are considered. Finite-resolution effects may therefore persist in weakly constrained or multi-solution regions. More importantly for experimental applications, real lidar observations are affected by continuous and structured measurement errors, including channel-dependent noise, calibration uncertainty, and correlated perturbations, which cannot be fully represented by a finite set of discrete lookup-table candidates. Thus, the response of LUT to observational perturbations is often characterized by locally piecewise or switching-like behavior, rather than by a fully smooth estimator response.

By contrast, for a continuous regression model trained under the dominance of squared loss, the optimal solution tends statistically toward the conditional expectation:

$$g^*(\mathbf{y}) = \mathbb{E}[\boldsymbol{\theta} \mid \mathbf{y}]. \tag{27}$$

This does not mean that the network explicitly computes an analytic posterior mean. Rather, under a continuous regression framework dominated by squared error, its output can be interpreted, in a statistical sense, as a mean-like compromise over the local conditional distribution. Therefore, when the local conditional distribution broadens from concentrated to

dispersed, the output of UNC usually remains near the region supported by the main probability mass and is less likely to exhibit abrupt switching induced by discrete candidate replacement.

This difference is illustrated intuitively in Fig. 6(a)–(c). For m_r, Fig. 6(a) shows that under the clean condition, the conditional distribution is relatively concentrated, and the true value, the LUT estimate, and the UNC estimate are all located near the main peak. Under the noisy condition, however, the distribution broadens significantly and shifts rightward. In this case, the LUT estimate is more likely to move away from the main high-density region, whereas UNC remains near the region supported by the dominant probability mass. This is consistent with the result in Section IV.A that the bias of m_r changes from negative to positive for LUT while UNC maintains a small negative bias. This suggests that m_r is better constrained by the optical observations than strongly ill-posed quantities such as N, but its estimate can still drift systematically when noise increases. In this case, the advantage of UNC is that its bias changes more slowly and remains in a small negative range, whereas the LUT estimate shows a steeper shift and changes from negative to positive bias.

For m_i, Fig. 6(b) shows that the conditional distribution remains concentrated in a narrow range close to zero. The main difference between the clean and noisy conditions is not a global shift of the distribution, but rather a reshaping of its form and a local broadening. Under such circumstances, the difference between the two methods is no longer primarily expressed as a clear sign reversal, but rather as a difference in how stable the estimate remains relative to the high-density region. Under noise, LUT is more likely to move toward the edge region, whereas UNC changes more continuously. This corresponds to the smoother bias evolution observed in Section IV.A, rather than to a simple absolute-error advantage.

The right-skewed tails in the empirical densities of m_r and m_i further indicate that, even when the dominant probability mass is concentrated near the true value, a set of higher-refractive-index solutions can still remain optically compatible with the observation within the local uncertainty range. This tail behavior is consistent with the existence of low-misfit solution valleys in ill-posed microphysical retrieval, where changes in the refractive index can be partially compensated by changes in the size-distribution parameters.

By contrast, the conditional distribution of N in Fig. 6(c) exhibits a more typical strongly ill-posed structure. Even under the clean condition, its distribution is already much broader than that of the refractive index parameters. After noise is added, it broadens further and becomes more skewed with a longer tail. This means that, for N, even a small perturbation in observation space may correspond to a set of candidate parameter states that differ substantially from one another. Under such conditions, LUT is more likely to jump toward a discrete candidate far from the central region, thereby exhibiting a rapid increase in error. UNC is also affected by the broadening of the distribution, but its output remains more strongly concentrated near the main probability mass region. Therefore, for a strongly ill-posed parameter of this kind, the advantage of deep learning lies primarily in delaying instability, rather than in eliminating the ill-posedness itself.

Thus, what Fig. 6(a)–(c) reveals is not a simple comparison of "which estimate is closer to a single peak," but rather the different response modes of the two estimators when faced with the broadening of the same local conditional distribution. LUT is more strongly governed by discrete candidate switching, whereas UNC is more inclined to produce a continuous statistical compromise located near the region supported by the main probability mass. For this reason, the advantage of deep learning lies not in increasing observational information, but in producing more continuous local responses under ill-posed conditions. This also lays the foundation for the next subsection, where the response geometry of the two estimators is visualized directly on a local perturbation plane.

C. Geometric response on the observation perturbation plane

The previous two subsections have shown that, under the same increase in noise, UNC tends to exhibit smoother bias trajectories and a slower degradation rate. The purpose of this subsection is therefore not to prove additionally that UNC is more robust to every possible perturbation. Rather, the goal is to explain directly from observation space why such smaller deviations arise in the local perturbation neighborhood considered here. One immediate reason is that the estimators have different response shapes to local observational variations. Accordingly, this subsection no longer examines the parameter distribution under fixed observations. Instead, it directly studies how the estimator $\hat{\theta}(y)$ varies when the observation y is continuously perturbed within a local neighborhood, that is, its local geometric response.

To this end, m_r is selected as a representative parameter, and the local response surfaces given by different methods are visualized on a two-dimensional perturbation plane. This plane should be understood as a representative local slice of the five-dimensional observation space, rather than as an exhaustive description of all possible perturbation directions. Similar response surfaces could in principle be constructed for other pairs of optical channels and other retrieved parameters, but such a full analysis would be difficult to present and interpret in the main text. Specifically, with the reference observation $\mathbf{y}_0$ from Section IV.B as the center, relative perturbations are applied only to two key optical quantities, β_{1064} and α_{532}. A local grid is therefore constructed along the directions $\Delta\beta_{1064}/\beta_{1064,0}$ and $\Delta\alpha_{532}/\alpha_{532,0}$, where the subscript 0 denotes the corresponding value in the reference observation $\mathbf{y}_0$. The remaining observational components are kept at the reference state or synchronized according to the prescribed rule.

For each perturbed observation point $\mathbf{y_0} + \boldsymbol{\delta}\mathbf{y}$ on this plane, the LUT and UNC estimators are evaluated separately to obtain the corresponding predicted value of m_r, and the signed relative error with respect to the truth is then constructed. Thus, what is shown in the figure is not a distribution of parameter samples, but rather the geometric projection of the mapping from "observation perturbation" to "parameter estimate" in a local observation neighborhood.

In this sense, Fig. 7 can be viewed as a geometric

complement to the mechanism of "discrete matching versus continuous statistical compromise" discussed in Section IV.B. If an estimator changes continuously under local perturbations, its response on the two-dimensional perturbation plane will generally appear as a smooth and traceable surface. By contrast, if it is highly sensitive to small perturbations and involves discrete switching, it is more likely to exhibit steps, folded boundaries, and local jumps.

As shown in Fig. 7, within the same perturbation range, the m_r surface corresponding to UNC is overall much smoother and varies continuously and progressively with the observations. By contrast, the LUT surface exhibits more pronounced segmented plateaus, folded boundaries, and local discontinuities. The error heatmaps above further provide quantitative evidence. Without additional shift, the Mean RE and Mean |RE| of UNC are +1.35% and 1.35%, respectively, whereas those of LUT reach +3.35% and 5.32%, which are about 2.48 and 3.94 times those of UNC. When the perturbation plane is additionally shifted along the α_{532} direction, the Mean RE and Mean |RE| of UNC are +3.11% and 3.12%, whereas LUT increases to +5.67% and 6.38%, corresponding to about 1.82 and 2.04 times those of UNC. When the perturbation plane is additionally shifted along the β_{1064} direction, the Mean RE of UNC is nearly unbiased at -0.09%, and Mean |RE| is 0.82%, whereas LUT gives -0.52% and 4.80%, respectively, with an average absolute relative error about 5.85 times that of UNC. These results indicate that the advantage of UNC is reflected not only in a smoother response surface, but also in a more uniform error field and a more concentrated low-error region, whereas LUT is more likely to form high-error bands near switching boundaries and to let them expand with perturbation.

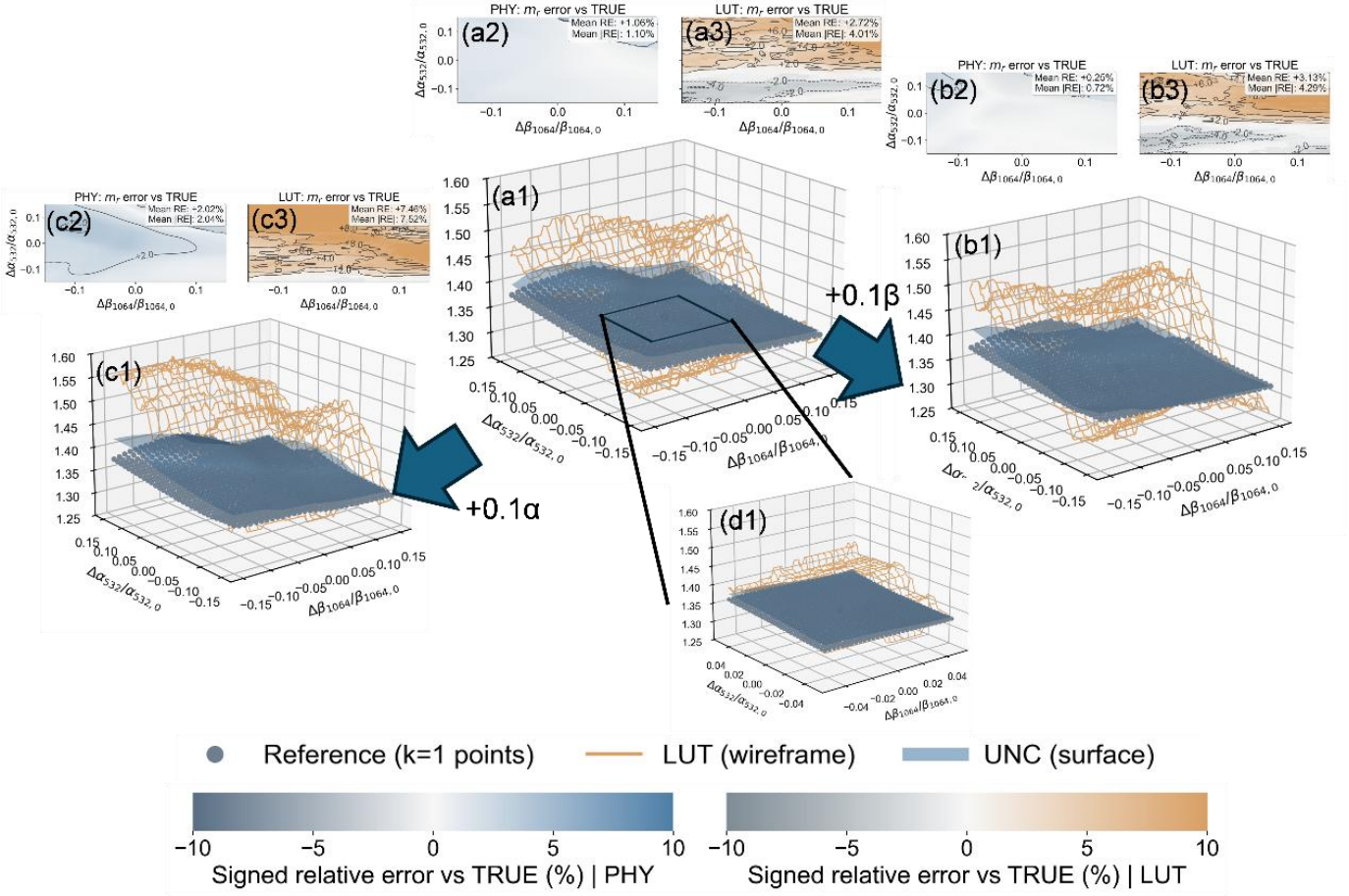


Fig. 7. Comparison of the $m_r$response surfaces on the original and additionally shifted local perturbation planes. Panel (a1) shows the m_r response surface on the original two-dimensional local perturbation plane defined by $\Delta\beta_{1064}/\beta_{1064,0}$ and $\Delta\alpha_{532}/\alpha_{532,0}$. The orange wireframe denotes LUT, the blue surface denotes UNC, and the gray-blue dots denote the local reference samples. Panels (a2) and (a3) show the signed relative error heatmaps of UNC and LUT with respect to the true value of the selected reference sample. Panels (b1)–(b3) correspond to the results under an additional shift along the β_{1064}direction, and panels (c1)–(c3) correspond to those under an additional shift along the α_{532}direction. Panel (d1) gives a zoomed-in view of the central region. The mean relative error (Mean RE) and mean absolute relative error (Mean | RE |) are annotated in each local region to compare the response continuity and error distribution characteristics of the two estimators under local perturbations. The perturbation axes are normalized by the corresponding values in the reference observation $\boldsymbol{y_0}$; for example, $\beta_{1064,0}$ and $\alpha_{532,0}$ denote the reference values of β_{1064} and α_{532}, respectively.

This difference is consistent with the results of the previous two subsections. In Section IV.A, m_r under LUT shows a negative-to-positive bias reversal and faster error expansion. In

Section IV.B, its empirical conditional density under the noisy condition broadens substantially and shifts rightward, making the LUT estimate more likely to jump into the right tail region, while UNC remains near the main probability mass region. Fig. 7 further reveals the local source of this phenomenon from the point of view of the observation space. The LUT response surface is assembled from multiple segmented regions; therefore, once a perturbation path crosses the boundary between them, the output may jump abruptly. By contrast, the UNC response surface is closer to a locally continuous surface; therefore, even when there is an overall shift, its variation trajectory remains comparatively smooth. Accordingly, the advantage of deep learning here does not lie in "creating new physical laws" or "obtaining more observational information," but in exhibiting fewer abrupt local responses caused by discrete candidate switching within the local observation neighborhood considered here, thereby providing a smoother and more stable point estimate for ill-posed retrieval.

This behavior has practical relevance for atmospheric remote-sensing retrieval. Within a relatively homogeneous aerosol layer and in the absence of evidence for a genuine regime transition, small and continuous variations in the optical observations should not be unnecessarily amplified into abrupt changes in the retrieved microphysical parameters. Such jumps may arise from discrete candidate switching and can complicate time–height statistics and subsequent interpretation. A locally continuous estimator response is therefore desirable as a measure of numerical stability. However, smoothness alone does not establish physical accuracy: it may also arise from regularization or conditional averaging, and excessive smoothing could obscure genuine aerosol boundaries or rapid microphysical changes. The local-response analysis must therefore be interpreted jointly with synthetic truth, in situ comparisons, and observation-space consistency.

It should be emphasized that the statement "the local response surface is smoother" is valid only for the current reference sample and perturbation neighborhood and should not be interpreted as a comprehensive robustness guarantee for all out-of-distribution samples, worst-case perturbations, or systematic bias scenarios. To avoid overgeneralization, Appendix C further provides lightweight sensitivity tests under several structured perturbations. Their results are consistent with the conclusion in the main text: under typical structured perturbations, UNC exhibits fewer locally abrupt responses than LUT.

V. Conclusion and Discussion

This study addressed aerosol microphysical retrieval from multiwavelength lidar as a representative ill-posed inverse problem. Building on AecroFormer, we developed a physics-informed learning framework that combines a group-normalized optical representation supplemented by two same-wavelength lidar-ratio features, observation-space physical consistency constraints, dynamic non-gridded sampling, heteroscedastic uncertainty modeling, and physically guided data augmentation. The goal was not to increase the amount of observational information, but to construct a more stable estimator under the same $3\beta + 2\alpha$ optical constraints.

The practical relevance of the present 3β+2α configuration should also be emphasized. This configuration, consisting of three particle backscatter coefficients and two particle extinction coefficients, is not only a theoretical test case but also closely related to the measurement capabilities of several advanced multiwavelength lidar systems, including CMA-operated CARLNET Raman lidar systems [55], many EARLINET/ACTRIS Raman lidar stations (https://www.earlinet.eu/), and the NASA Langley airborne HSRL-2 system (https://airbornescience.nasa.gov/instrument/High_Spectral_Resolution_Lidar_2). Therefore, there is a clear need for retrieval schemes that can robustly convert lidar-derived aerosol optical products into physically meaningful aerosol microphysical parameters under real observational conditions.

The synthetic ablation experiments showed that the proposed components play complementary and parameter-dependent roles rather than producing a uniformly progressive improvement in every metric. The reorganized input representation provided the most consistent improvement in point-estimate accuracy relative to the original AecroFormer. The physical consistency loss improved selected parameter-bias structures and reduced the overall observation-space reconstruction error for the DISCOVER-AQ observations, although it did not uniformly improve every microphysical metric or optical channel. Its role is therefore interpreted primarily as observation-space regularization within the adopted Lorenz–Mie forward model. Dynamic non-gridded sampling reduced dependence on a fixed discrete training distribution, although its independent effect on average point-estimate accuracy was modest. The uncertainty-enabled configuration maintained competitive point-estimate accuracy and provided parameter-dependent predictive intervals. The strong degradation and increased inter-run variability observed after removing data augmentation further confirmed the importance of physically guided perturbation design for robustness. These results indicate that retrieval accuracy, stability, and credibility arise from the complementary effects of improved input representation, continuous parameter-space sampling, observation-space consistency, structured error augmentation, and uncertainty-aware supervision, rather than from a single uniformly dominant module. A CPU-based timing benchmark further showed that the uncertainty-aware configuration reduced the retrieval time from 1.94 s to 0.022 ms per range-resolved sample compared with FAST, corresponding to an acceleration of approximately 8.8×10^4 for the same microphysical inversion task.

External validation with NASA DISCOVER-AQ California in situ observations further supported the applicability of the framework under selected real observational conditions. Based on 30 valid matched profiles from 8 observation days and 14 B-200 flight segments, BAS-D achieved an r_e MARE of 14.36%±0.86%, equal to the LUT (FAST) value of 14.36%. UNC yielded a comparable r_e MARE of 14.59%±1.67%,

while its signed bias of -3.99%±1.26% was smaller in magnitude than the LUT bias of -10.92%. For SSA_{532}, DYN and UNC achieved MAE values of 0.0225±0.0032 and 0.0247±0.0032, respectively, both lower than the LUT value of 0.0370. However, these results do not represent a uniform improvement over ORI, which already achieved an SSA_{532} MAE of 0.0223±0.0009. The field results therefore reveal parameter-dependent trade-offs rather than universal superiority of a single configuration.

The 1 February 2013 case further showed that UNC produced more continuous time–height and vertical-profile responses under realistic aerosol variability while remaining locally competitive with LUT for particle-size-related quantities. The model-derived 68% predictive intervals additionally identified altitude ranges associated with larger retrieval uncertainty. These intervals support uncertainty-aware interpretation of the profiles but do not include errors associated with aircraft collocation, vertical averaging, or the in-situ reference and should not be interpreted as strictly calibrated field confidence intervals. The observed continuity and interval behavior characterize estimator stability and credibility rather than independently verified physical truth.

The present external validation remains limited to 30 matched profiles collected during 8 DISCOVER-AQ observation days and 14 B-200 flight segments. The available sample size does not support reliable simultaneous stratification by date, aerosol loading, relative humidity, altitude, and aerosol type. Broader validation will require additional independent campaigns with collocated multiwavelength lidar and in-situ microphysical observations, together with extensions of the forward model for coarse-mode, multimodal, and nonspherical aerosol populations. Future work will extend the evaluation to independent airborne campaigns, such as ORACLES for biomass-burning aerosol and, after appropriate forward-model extensions, ACTIVATE for more complex marine and mixed aerosol conditions.

The perturbation and noise analyses clarified why deep learning can be advantageous in this ill-posed setting. Different parameters showed different degradation modes: m_r was mainly affected by noise-induced bias drift, m_i by changes in the local statistical compromise, and N by strong ill-posed amplification. The conditional-distribution and local perturbation-plane analyses further showed that LUT is more sensitive to discrete candidate switching, whereas UNC tends to produce smoother local responses near the dominant compatible solution region. Therefore, the advantage of deep learning should be understood as showing a smoother estimator behavior, slower degradation, and fewer abrupt local responses, rather than as a fundamental relaxation of the information limit imposed by the observations.

This interpretation of the results also defines the boundary of the proposed framework. Because the present forward operator is based on spherical Lorenz–Mie scattering, strongly nonspherical particles may introduce systematic forward-model mismatch, particularly for aerosol populations dominated by mineral dust or other irregular particles. Under such conditions, the estimator may return an effective spherical solution that reproduces part of the measured optical response but does not represent the true particle shape or microphysical state. Extension to nonspherical aerosol conditions would require appropriate nonspherical scattering kernels, possible incorporation of depolarization information, regeneration of the training data, and independent validation. The conclusions drawn in this study apply specifically to the fine-mode-dominated unimodal retrieval setting defined in Section II.A and should not be generalized directly to coarse-mode-dominated or strongly multimodal aerosol populations. Ill-posedness cannot be eliminated simply by increasing model complexity, and no learning method can recover information that is absent from the optical observations. Therefore, the role of deep learning in this context is not to remove the fundamental non-uniqueness of the inverse problem, but to construct a more stable estimator by integrating representative uncertainty structures and physical constraints into the learning process. Reliable calibration and high-quality optical inputs remain essential. If one or more lidar channels contain unnoticed calibration biases or processing-related offsets, a learning-based estimator may still generate a smooth solution that is internally consistent with biased observations but physically inaccurate. Future work should therefore examine calibration-bias sensitivity, channel-dependent systematic errors, and out-of-distribution optical inputs, and should combine retrieval models with independent calibration checks or consistency diagnostics.

More broadly, this work suggests that AI-based retrieval and observing-system development should be viewed as complementary. Deep learning can improve how existing observational information is used, but the upper limit of recoverable microphysical information is still determined by the forward physics, instrumental quality, and observational constraints. Future extensions should incorporate more realistic forward models, including nonspherical particles, bimodal or multimodal size distributions, core–shell particle-mixing structures, and wavelength- and size-dependent refractive indices. Another promising direction is to move from point-wise retrieval toward segment- or curtain-plot-level retrieval, where vertical and temporal coherence of aerosol fields can be used as additional regularization information.

Overall, the main contribution of this study is not a new network backbone, but a physically constrained, uncertainty-aware, and mechanism-interpretable learning framework for more stable aerosol microphysical retrieval under ill-posed multiwavelength lidar conditions.

Data availability

The code and pretrained models will be made publicly available at https://github.com/Wiki-Chou/PI_AecroFormer.

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